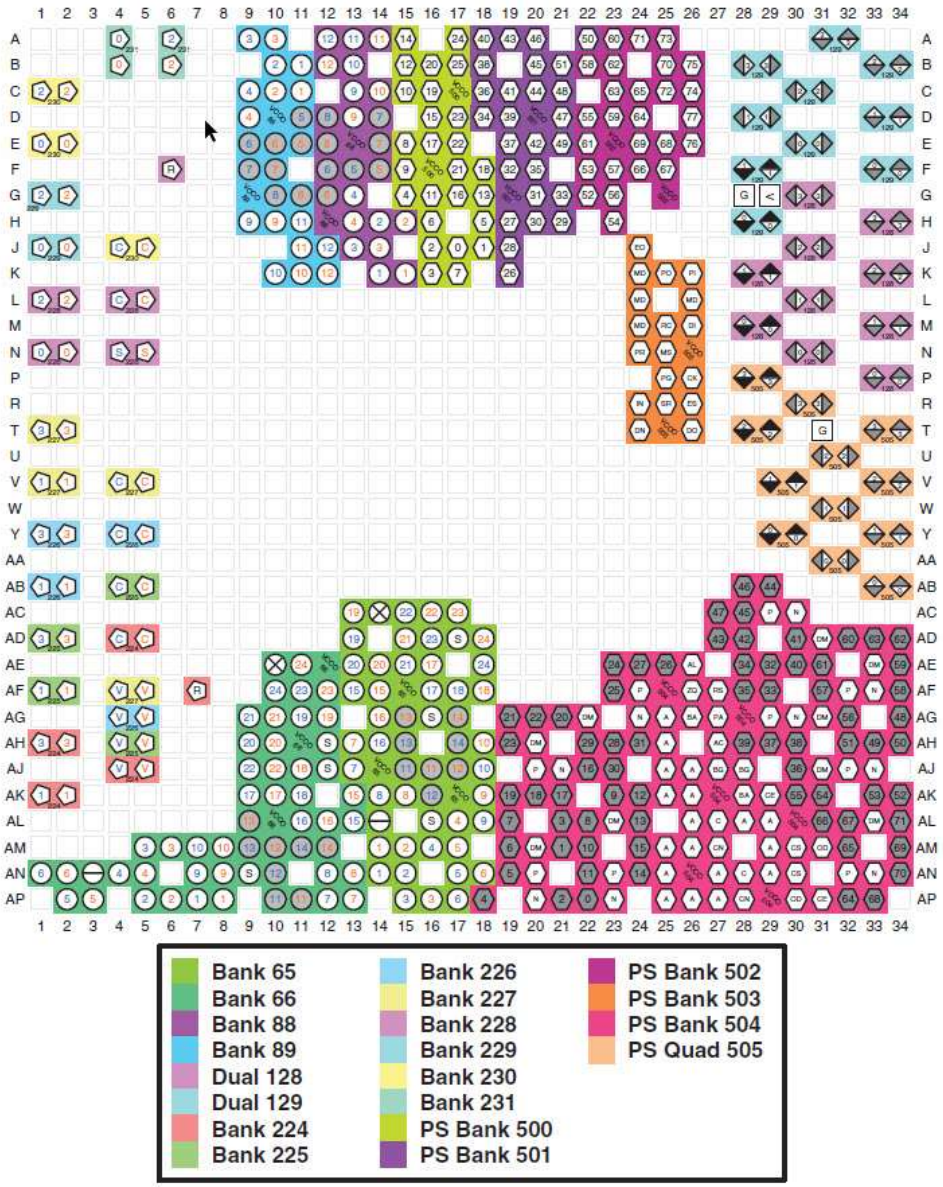
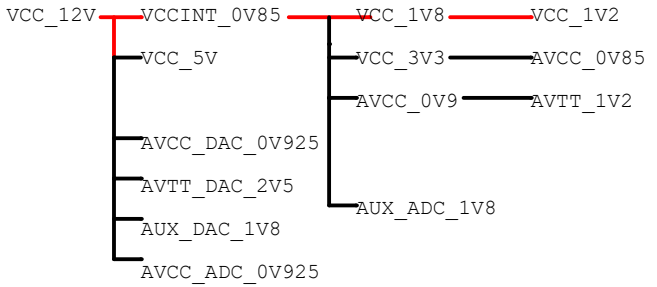
项目信息	项目名称	RF_CORE_BOARD	板卡设计者	hanxudong
	PCB板名	RF_CORE_BOARD	PCB版本号	1.3
	项目情况	☑正常项目 ☐紧急项目，原因：		
基本信息	PCB层数	14 层	工艺要求	☐喷锡 ☐沉金 ☑ _____
	PCB板厚	2.0 mm	阻焊要求	☐双面红油 ☑其它：_____
	加工数量	? 块	钢网加工	☐需要 ☐不需要
	项目负责人		软件开发人员	
设计要求	电路要求	1、原理图是否按设计规范检查？ ☑是 ☐否 2、原理图是否有使用新器件？ ☑无 ☐有，请在下框列出新器件名称/位号 3、原理图是否为全新设计？ ☐是 ☑否，请在下框里写上原理图修改记录、修改人和修改日期 修改IIC线序 4、软件是否已核对管脚分配？ ☑是 ☐否，原因： 确认人员： 5、DRC检查结果Error/Warning是否为0？ ☑是 ☐否 6、其他特殊布线要求说明：		
	结构要求	1、PCB结构定位是否已确认？ ☑是 ☐否，原因： 确认人员： 2、是否需要设计结构样板？ ☐是 ☑否 ，结构其他说明：		
基他要求		1、加工周期： ☑普通 ☐加急 2、计划投板周期：		

	V3BEST	
Title	RF_CORE_BOARD	
Size A4	Document Number 00 PCB DESIGNER	Rev V1.3
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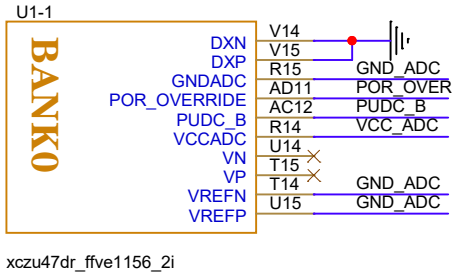


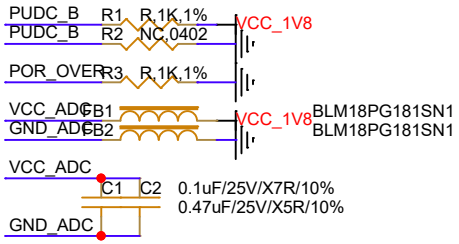


P00_PCB_DESIGNER
P01_SYSTEM_DIAGRAM
P02_FPGA_BANK0
P03_FPGA_BANK_65-66
P04_FPGA_BANK_88-89
P05_FPGA_BANK_128_129
P06_FPGA_BANK_224-227
P07_FPGA_BANK_228-231
P08_FPGA_BANK_500-502
P09_FPGA_BANK_503
P10_FPGA_BANK_504
P11_FPGA_BANK_505
P12_FPGA_ANALOG_POWER
P13_FPGA_POWER_GTY_AUX
P14_FPGA_POWER_INT
P15_FPGA_POWER_PS
P16_FPGA_GND
P17_PL_DDR4
P18_PL_DDR4_TERMINAL
P19_PS_DDR4_00
P20_PS_DDR4_01
P21_PS_DDR4_TERMINAL
P22_MEMORY
P23_CLOCK
P24_SYS_CLOCK
P25_POWER_VCCINT00
P26_POWER_VCCINT01
P27_POWER_VCCINT_0V9_3V3_1V2
P28_POWER_VCC_1V8_2V2
P29_RF_POWER00
P30_RF_POWER01
P31_CONNECTOR_ANALOG
P32_CONNECTOR_DIGTAL00
P33_CONNECTOR_DIGITAL01

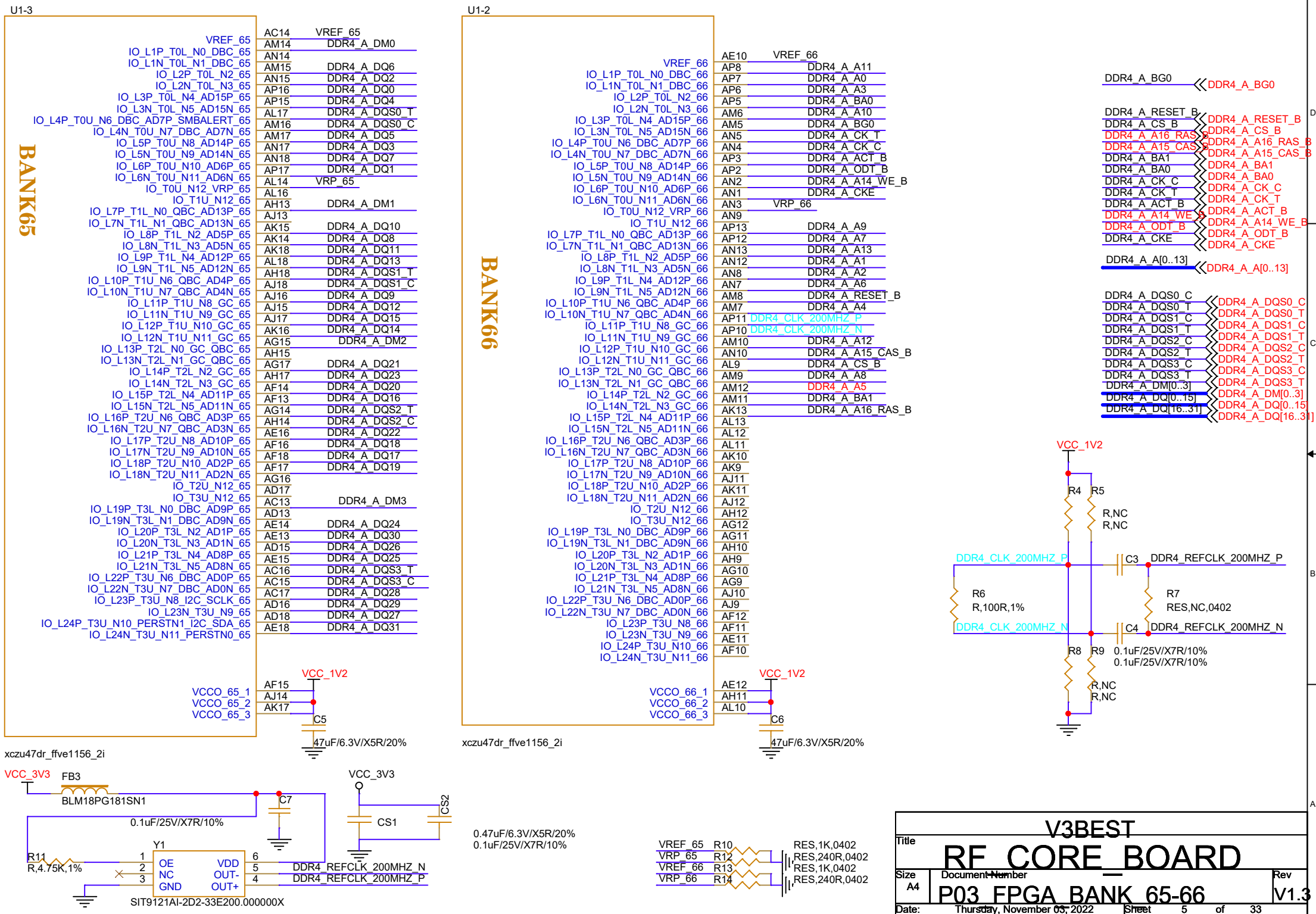
RF_SOM_CORE_BOARD

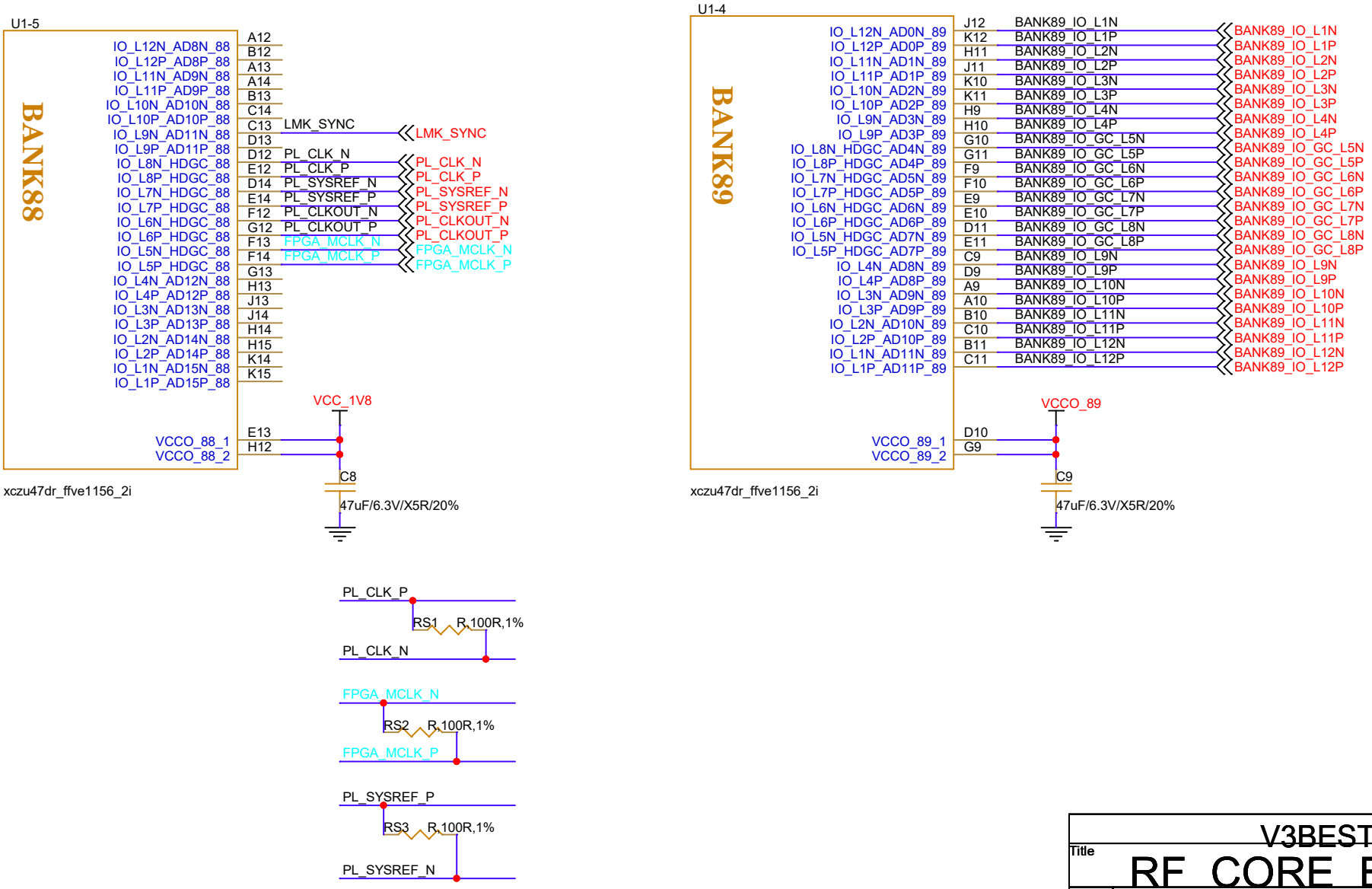
Title	V3BEST RF_CORE_BOARD		
Size A3	Document Number P01_SYSTEM_DIAGRAM		Rev V1.3
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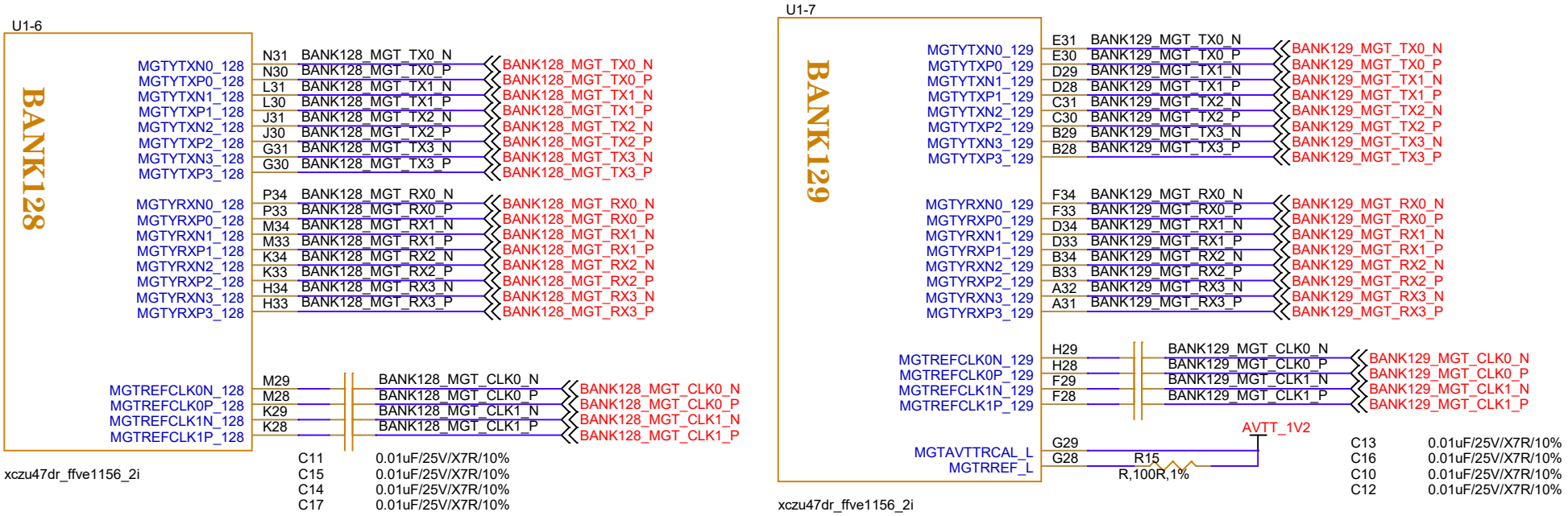


	V3BEST	
Title	RF_CORE_BOARD	
Size A4	Document Number P02_FPGA_BANK0	Rev V1.3
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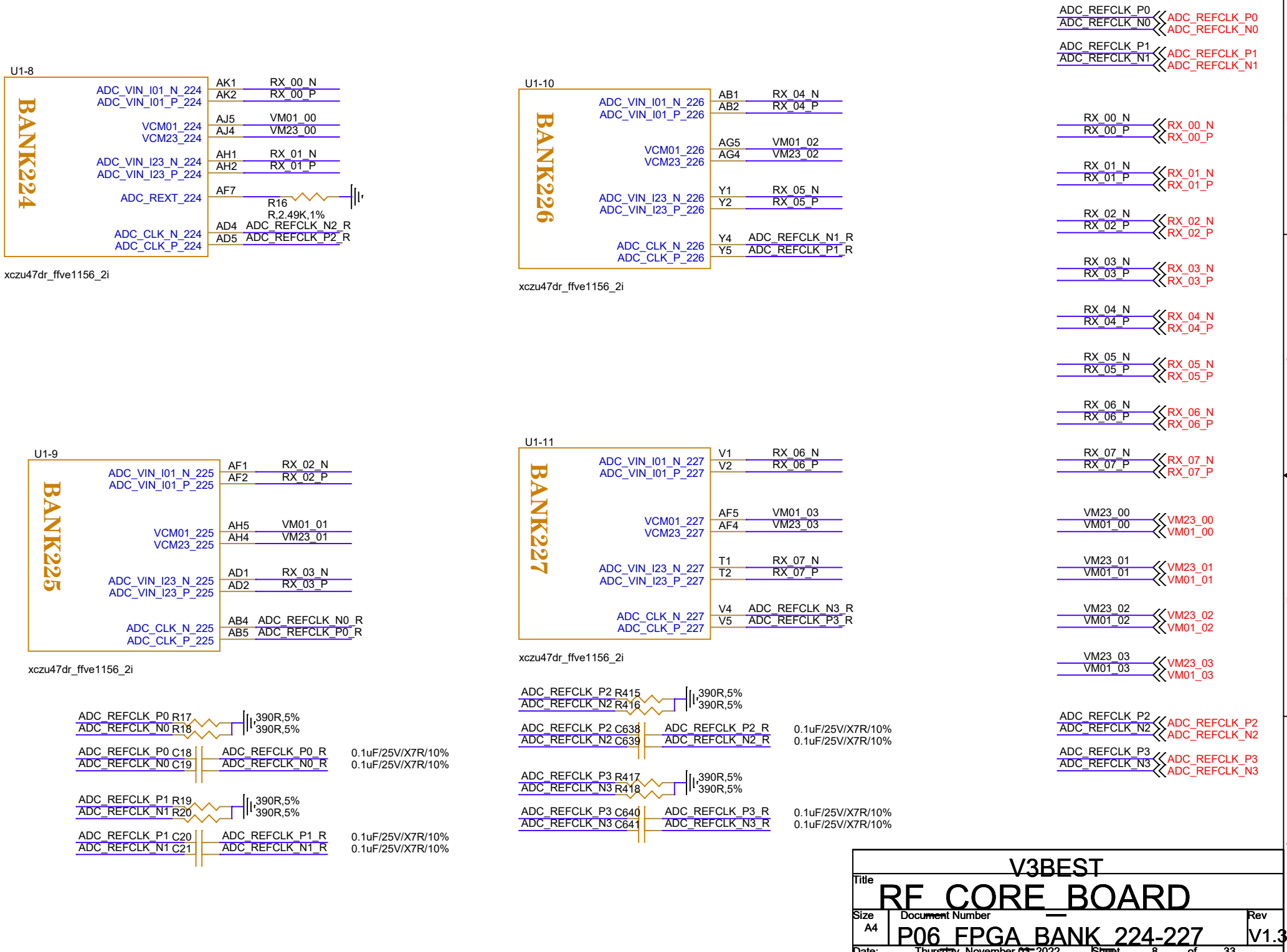

U1-3
BANK65
VREF_65
IO_L1P_T0L_N0_DBC_65
IO_L1N_T0L_N1_DBC_65
IO_L2P_T0L_N2_65
IO_L2N_T0L_N3_65
IO_L3P_T0L_N4_AD15P_65
IO_L3N_T0L_N5_AD15N_65
IO_L4P_T0U_N6_DBC_AD7P_SMBALERT_65
IO_L4N_T0U_N7_DBC_AD7N_65
IO_L5P_T0U_N8_AD14P_65
IO_L5N_T0U_N9_AD14N_65
IO_L6P_T0U_N10_AD6P_65
IO_L6N_T0U_N11_AD6N_65
IO_T0U_N12_VRP_65
IO_T1U_N12_65
IO_L7P_T1L_N0_QBC_AD13P_65
IO_L7N_T1L_N1_QBC_AD13N_65
IO_L8P_T1L_N2_AD5P_65
IO_L8N_T1L_N3_AD5N_65
IO_L9P_T1L_N4_AD12P_65
IO_L9N_T1L_N5_AD12N_65
IO_L10P_T1U_N6_QBC_AD4P_65
IO_L10N_T1U_N7_QBC_AD4N_65
IO_L11P_T1U_N8_GC_65
IO_L11N_T1U_N9_GC_65
IO_L12P_T1U_N10_GC_65
IO_L12N_T1U_N11_GC_65
IO_L13P_T2L_N0_GC_QBC_65
IO_L13N_T2L_N1_GC_QBC_65
IO_L14P_T2L_N2_GC_65
IO_L14N_T2L_N3_GC_65
IO_L15P_T2L_N4_AD11P_65
IO_L15N_T2L_N5_AD11N_65
IO_L16P_T2U_N6_QBC_AD3P_65
IO_L16N_T2U_N7_QBC_AD3N_65
IO_L17P_T2U_N8_AD10P_65
IO_L17N_T2U_N9_AD10N_65
IO_L18P_T2U_N10_AD2P_65
IO_L18N_T2U_N11_AD2N_65
IO_T2U_N12_65
IO_T3U_N12_65
IO_L19P_T3L_N0_DBC_AD9P_65
IO_L19N_T3L_N1_DBC_AD9N_65
IO_L20P_T3L_N2_AD1P_65
IO_L20N_T3L_N3_AD1N_65
IO_L21P_T3L_N4_AD8P_65
IO_L21N_T3L_N5_AD8N_65
IO_L22P_T3U_N6_DBC_AD0P_65
IO_L22N_T3U_N7_DBC_AD0N_65
IO_L23P_T3U_N8_I2C_SCLK_65
IO_L23N_T3U_N9_65
IO_L24P_T3U_N10_PERSTN1_I2C_SDA_65
IO_L24N_T3U_N11_PERSTN0_65
AC14 VREF_65
AM14 DDR4_A_DM0
AN14
AM15 DDR4_A_DQ6
AN15 DDR4_A_DQ2
AP16 DDR4_A_DQ0
AP15 DDR4_A_DQ4
AL17 DDR4_A_DQS0_T
AM16 DDR4_A_DQS0_C
AM17 DDR4_A_DQ5
AN17 DDR4_A_DQ3
AN18 DDR4_A_DQ7
AP17 DDR4_A_DQ1
AL14 VRP_65
AL16
AH13 DDR4_A_DM1
AJ13
AK15 DDR4_A_DQ10
AK14 DDR4_A_DQ8
AK18 DDR4_A_DQ11
AL18 DDR4_A_DQ13
AH18 DDR4_A_DQS1_T
AJ18 DDR4_A_DQS1_C
AJ16 DDR4_A_DQ9
AJ15 DDR4_A_DQ12
AJ17 DDR4_A_DQ15
AK16 DDR4_A_DQ14
AG15 DDR4_A_DM2
AH15
AG17 DDR4_A_DQ21
AH17 DDR4_A_DQ23
AF14 DDR4_A_DQ20
AF13 DDR4_A_DQ16
AG14 DDR4_A_DQS2_T
AH14 DDR4_A_DQS2_C
AE16 DDR4_A_DQ22
AF16 DDR4_A_DQ18
AF18 DDR4_A_DQ17
AF17 DDR4_A_DQ19
AG16
AD17
AC13 DDR4_A_DM3
AD13
AE14 DDR4_A_DQ24
AE13 DDR4_A_DQ30
AD15 DDR4_A_DQ26
AE15 DDR4_A_DQ25
AC16 DDR4_A_DQS3_T
AC15 DDR4_A_DQS3_C
AC17 DDR4_A_DQ28
AD16 DDR4_A_DQ29
AD18 DDR4_A_DQ27
AE18 DDR4_A_DQ31
VCC_1V2
AF15
AJ14
AK17
VCCO_65_1
VCCO_65_2
VCCO_65_3
C5
47uF/6.3V/X5R/20%
xczu47dr_ffve1156_2i
U1-2
BANK66
VREF_66
IO_L1P_T0L_N0_DBC_66
IO_L1N_T0L_N1_DBC_66
IO_L2P_T0L_N2_66
IO_L2N_T0L_N3_66
IO_L3P_T0L_N4_AD15P_66
IO_L3N_T0L_N5_AD15N_66
IO_L4P_T0U_N6_DBC_AD7P_66
IO_L4N_T0U_N7_DBC_AD7N_66
IO_L5P_T0U_N8_AD14P_66
IO_L5N_T0U_N9_AD14N_66
IO_L6P_T0U_N10_AD6P_66
IO_L6N_T0U_N11_AD6N_66
IO_T0U_N12_VRP_66
IO_T1U_N12_66
IO_L7P_T1L_N0_QBC_AD13P_66
IO_L7N_T1L_N1_QBC_AD13N_66
IO_L8P_T1L_N2_AD5P_66
IO_L8N_T1L_N3_AD5N_66
IO_L9P_T1L_N4_AD12P_66
IO_L9N_T1L_N5_AD12N_66
IO_L10P_T1U_N6_QBC_AD4P_66
IO_L10N_T1U_N7_QBC_AD4N_66
IO_L11P_T1U_N8_GC_66
IO_L11N_T1U_N9_GC_66
IO_L12P_T1U_N10_GC_66
IO_L12N_T1U_N11_GC_66
IO_L13P_T2L_N0_GC_QBC_66
IO_L13N_T2L_N1_GC_QBC_66
IO_L14P_T2L_N2_GC_66
IO_L14N_T2L_N3_GC_66
IO_L15P_T2L_N4_AD11P_66
IO_L15N_T2L_N5_AD11N_66
IO_L16P_T2U_N6_QBC_AD3P_66
IO_L16N_T2U_N7_QBC_AD3N_66
IO_L17P_T2U_N8_AD10P_66
IO_L17N_T2U_N9_AD10N_66
IO_L18P_T2U_N10_AD2P_66
IO_L18N_T2U_N11_AD2N_66
IO_T2U_N12_66
IO_T3U_N12_66
IO_L19P_T3L_N0_DBC_AD9P_66
IO_L19N_T3L_N1_DBC_AD9N_66
IO_L20P_T3L_N2_AD1P_66
IO_L20N_T3L_N3_AD1N_66
IO_L21P_T3L_N4_AD8P_66
IO_L21N_T3L_N5_AD8N_66
IO_L22P_T3U_N6_DBC_AD0P_66
IO_L22N_T3U_N7_DBC_AD0N_66
IO_L23P_T3U_N8_66
IO_L23N_T3U_N9_66
IO_L24P_T3U_N10_66
IO_L24N_T3U_N11_66
AE10 VREF_66
AP8 DDR4_A_A11
AP7 DDR4_A_A0
AP6 DDR4_A_A3
AP5 DDR4_A_BA0
AM6 DDR4_A_A10
AM5 DDR4_A_BG0
AN5 DDR4_A_CK_T
AN4 DDR4_A_CK_C
AP3 DDR4_A_ACT_B
AP2 DDR4_A_ODT_B
AN2 DDR4_A_A14_WE_B
AN1 DDR4_A_CKE
AN3 VRP_66
AN9
AP13 DDR4_A_A9
AP12 DDR4_A_A7
AN13 DDR4_A_A13
AN12 DDR4_A_A1
AN8 DDR4_A_A2
AN7 DDR4_A_A6
AM8 DDR4_A_RESET_B
AM7 DDR4_A_A4
AP11 DDR4_CLK_200MHZ_P
AP10 DDR4_CLK_200MHZ_N
AM10 DDR4_A_A12
AN10 DDR4_A_A15_CAS_B
AL9 DDR4_A_CS_B
AM9 DDR4_A_A8
AM12 DDR4_A_A5
AM11 DDR4_A_BA1
AK13 DDR4_A_A16_RAS_B
AL13
AL12
AL11
AK10
AK9
AJ11
AK11
AJ12
AH12
AG12
AG11
AH10
AH9
AG10
AG9
AJ10
AJ9
AF12
AF11
AE11
AF10
VCC_1V2
AE12
AH11
AL10
VCCO_66_1
VCCO_66_2
VCCO_66_3
C6
47uF/6.3V/X5R/20%
xczu47dr_ffve1156_2i
DDR4_A_BG0
DDR4_A_RESET_B
DDR4_A_CS_B
DDR4_A_A16_RAS_B
DDR4_A_A15_CAS_B
DDR4_A_BA1
DDR4_A_BA0
DDR4_A_CK_C
DDR4_A_CK_T
DDR4_A_ACT_B
DDR4_A_A14_WE_B
DDR4_A_ODT_B
DDR4_A_CKE
DDR4_A_A[0..13]
DDR4_A_DQS0_C
DDR4_A_DQS0_T
DDR4_A_DQS1_C
DDR4_A_DQS1_T
DDR4_A_DQS2_C
DDR4_A_DQS2_T
DDR4_A_DQS3_C
DDR4_A_DQS3_T
DDR4_A_DM[0..3]
DDR4_A_DQ[0..15]
DDR4_A_DQ[16..31]
VCC_1V2
R4 R5
R,NC
R,NC
DDR4_CLK_200MHZ_P
C3
DDR4_REFCLK_200MHZ_P
R6
R,100R,1%
R7
RES,NC,0402
DDR4_CLK_200MHZ_N
C4
DDR4_REFCLK_200MHZ_N
R8 R9
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
R,NC
R,NC
VCC_3V3
FB3
BLM18PG181SN1
C7
0.1uF/25V/X7R/10%
VCC_3V3
CS1
CS2
0.47uF/6.3V/X5R/20%
0.1uF/25V/X7R/10%
Y1
R11
R,4.75K,1%
OE
NC
GND
VDD
OUT-
OUT+
DDR4_REFCLK_200MHZ_N
DDR4_REFCLK_200MHZ_P
SIT9121AI-2D2-33E200.000000X
VREF_65 R10
VRP_65 R12
VREF_66 R13
VRP_66 R14
RES,1K,0402
RES,240R,0402
RES,1K,0402
RES,240R,0402
V3BEST
Title
RF_CORE_BOARD
Size A4
Document Number
P03_FPGA_BANK_65-66
Rev V1.3
Date: Thursday, November 03, 2022
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Title	V3BEST RF_CORE_BOARD	
Size: A4	Document Number: P04 FPGA_BANK_88-89	Rev: V1.3
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V3BEST		
Title: RF_CORE_BOARD		
Size: A4	Document Number: P05 FPGA_BANK_128_129	Rev: V1.3
Date: Thursday, November 03, 2022	Sheet 7 of 33	


U1-8
BANK224
ADC_VIN_I01_N_224
ADC_VIN_I01_P_224
VCM01_224
VCM23_224
ADC_VIN_I23_N_224
ADC_VIN_I23_P_224
ADC_REXT_224
ADC_CLK_N_224
ADC_CLK_P_224
AK1 RX_00_N
AK2 RX_00_P
AJ5 VM01_00
AJ4 VM23_00
AH1 RX_01_N
AH2 RX_01_P
AF7
R16
R,2.49K,1%
AD4 ADC_REFCLK_N2_R
AD5 ADC_REFCLK_P2_R
xczu47dr_ffve1156_2i
U1-10
BANK226
ADC_VIN_I01_N_226
ADC_VIN_I01_P_226
VCM01_226
VCM23_226
ADC_VIN_I23_N_226
ADC_VIN_I23_P_226
ADC_CLK_N_226
ADC_CLK_P_226
AB1 RX_04_N
AB2 RX_04_P
AG5 VM01_02
AG4 VM23_02
Y1 RX_05_N
Y2 RX_05_P
Y4 ADC_REFCLK_N1_R
Y5 ADC_REFCLK_P1_R
xczu47dr_ffve1156_2i
U1-9
BANK225
ADC_VIN_I01_N_225
ADC_VIN_I01_P_225
VCM01_225
VCM23_225
ADC_VIN_I23_N_225
ADC_VIN_I23_P_225
ADC_CLK_N_225
ADC_CLK_P_225
AF1 RX_02_N
AF2 RX_02_P
AH5 VM01_01
AH4 VM23_01
AD1 RX_03_N
AD2 RX_03_P
AB4 ADC_REFCLK_N0_R
AB5 ADC_REFCLK_P0_R
xczu47dr_ffve1156_2i
U1-11
BANK227
ADC_VIN_I01_N_227
ADC_VIN_I01_P_227
VCM01_227
VCM23_227
ADC_VIN_I23_N_227
ADC_VIN_I23_P_227
ADC_CLK_N_227
ADC_CLK_P_227
V1 RX_06_N
V2 RX_06_P
AF5 VM01_03
AF4 VM23_03
T1 RX_07_N
T2 RX_07_P
V4 ADC_REFCLK_N3_R
V5 ADC_REFCLK_P3_R
xczu47dr_ffve1156_2i
ADC_REFCLK_P0 R17
ADC_REFCLK_N0 R18
390R,5%
390R,5%
ADC_REFCLK_P0 C18
ADC_REFCLK_N0 C19
ADC_REFCLK_P0_R
ADC_REFCLK_N0_R
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
ADC_REFCLK_P1 R19
ADC_REFCLK_N1 R20
390R,5%
390R,5%
ADC_REFCLK_P1 C20
ADC_REFCLK_N1 C21
ADC_REFCLK_P1_R
ADC_REFCLK_N1_R
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
ADC_REFCLK_P2 R415
ADC_REFCLK_N2 R416
390R,5%
390R,5%
ADC_REFCLK_P2 C638
ADC_REFCLK_N2 C639
ADC_REFCLK_P2_R
ADC_REFCLK_N2_R
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
ADC_REFCLK_P3 R417
ADC_REFCLK_N3 R418
390R,5%
390R,5%
ADC_REFCLK_P3 C640
ADC_REFCLK_N3 C641
ADC_REFCLK_P3_R
ADC_REFCLK_N3_R
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
ADC_REFCLK_P0
ADC_REFCLK_N0
ADC_REFCLK_P0
ADC_REFCLK_N0
ADC_REFCLK_P1
ADC_REFCLK_N1
ADC_REFCLK_P1
ADC_REFCLK_N1
RX_00_N
RX_00_P
RX_00_N
RX_00_P
RX_01_N
RX_01_P
RX_01_N
RX_01_P
RX_02_N
RX_02_P
RX_02_N
RX_02_P
RX_03_N
RX_03_P
RX_03_N
RX_03_P
RX_04_N
RX_04_P
RX_04_N
RX_04_P
RX_05_N
RX_05_P
RX_05_N
RX_05_P
RX_06_N
RX_06_P
RX_06_N
RX_06_P
RX_07_N
RX_07_P
RX_07_N
RX_07_P
VM23_00
VM01_00
VM23_00
VM01_00
VM23_01
VM01_01
VM23_01
VM01_01
VM23_02
VM01_02
VM23_02
VM01_02
VM23_03
VM01_03
VM23_03
VM01_03
ADC_REFCLK_P2
ADC_REFCLK_N2
ADC_REFCLK_P2
ADC_REFCLK_N2
ADC_REFCLK_P3
ADC_REFCLK_N3
ADC_REFCLK_P3
ADC_REFCLK_N3
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P06_FPGA_BANK_224-227
Rev
V1.3
Date:
Thursday, November 03, 2022
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U1-12

BANK228

DAC_VOUT0_N_228 — N1 — TX_00_N
DAC_VOUT0_P_228 — N2 — TX_00_P

DAC_VOUT2_N_228 — L1 — TX_01_N
DAC_VOUT2_P_228 — L2 — TX_01_P

SYSREF_N_228 — N4 — DAC_SYSREF_N
SYSREF_P_228 — N5 — DAC_SYSREF_P

DAC_REXT_228 — F6 — R21 R,2.49K,1%

DAC_CLK_N_228 — L4 — DAC_REFCLK_N1_R
DAC_CLK_P_228 — L5 — DAC_REFCLK_P1_R

xczu47dr_ffve1156_2i

U1-13

BANK229

DAC_VOUT0_N_229 — J1 — TX_02_N
DAC_VOUT0_P_229 — J2 — TX_02_P

DAC_VOUT2_N_229 — G1 — TX_03_N
DAC_VOUT2_P_229 — G2 — TX_03_P

xczu47dr_ffve1156_2i

U1-14

BANK230

DAC_VOUT0_N_230 — E1 — TX_04_N
DAC_VOUT0_P_230 — E2 — TX_04_P

DAC_VOUT2_N_230 — C1 — TX_05_N
DAC_VOUT2_P_230 — C2 — TX_05_P

DAC_CLK_N_230 — J4 — DAC_REFCLK_N0_R
DAC_CLK_P_230 — J5 — DAC_REFCLK_P0_R

xczu47dr_ffve1156_2i

U1-15

BANK231

DAC_VOUT0_N_231 — A4 — TX_06_N
DAC_VOUT0_P_231 — B4 — TX_06_P

DAC_VOUT2_N_231 — A6 — TX_07_N
DAC_VOUT2_P_231 — B6 — TX_07_P

xczu47dr_ffve1156_2i

DAC_REFCLK_P0 — DAC_REFCLK_P0
DAC_REFCLK_N0 — DAC_REFCLK_N0

DAC_REFCLK_P1 — DAC_REFCLK_P1
DAC_REFCLK_N1 — DAC_REFCLK_N1

DAC_SYSREF_P — DAC_SYSREF_P
DAC_SYSREF_N — DAC_SYSREF_N

TX_00_N — TX_00_N
TX_00_P — TX_00_P

TX_01_N — TX_01_N
TX_01_P — TX_01_P

TX_02_N — TX_02_N
TX_02_P — TX_02_P

TX_03_N — TX_03_N
TX_03_P — TX_03_P

TX_04_N — TX_04_N
TX_04_P — TX_04_P

TX_05_N — TX_05_N
TX_05_P — TX_05_P

TX_06_N — TX_06_N
TX_06_P — TX_06_P

TX_07_N — TX_07_N
TX_07_P — TX_07_P

DAC_REFCLK_P1 R22 — 390R,5%
DAC_REFCLK_N1 R23 — 390R,5%

DAC_REFCLK_P1 C22 — DAC_REFCLK_P1_R — 0.1uF/25V/X7R/10%
DAC_REFCLK_N1 C23 — DAC_REFCLK_N1_R — 0.1uF/25V/X7R/10%

DAC_REFCLK_P0 R24 — 390R,5%
DAC_REFCLK_N0 R25 — 390R,5%

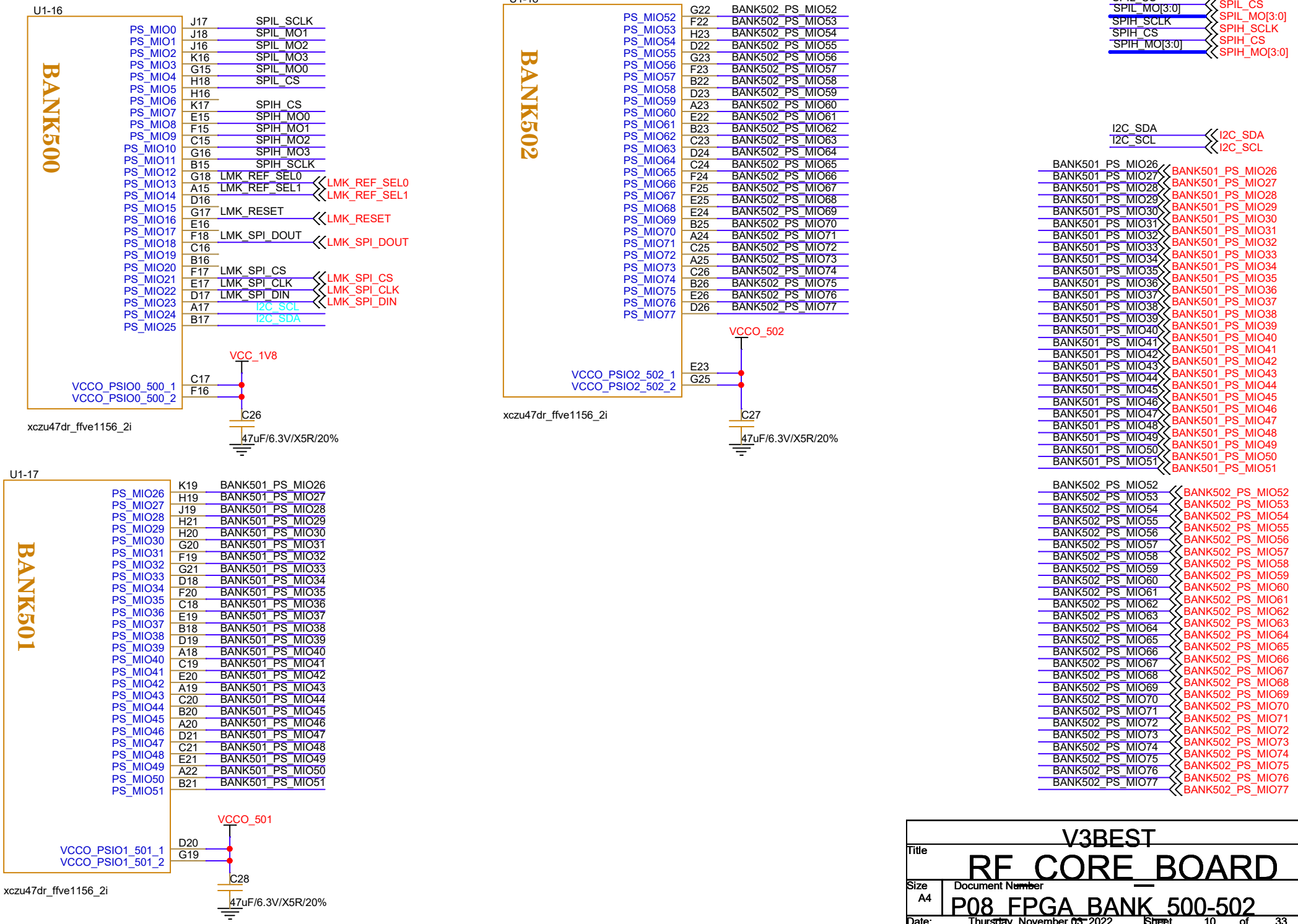
DAC_REFCLK_P0 C24 — DAC_REFCLK_P0_R — 0.1uF/25V/X7R/10%
DAC_REFCLK_N0 C25 — DAC_REFCLK_N0_R — 0.1uF/25V/X7R/10%

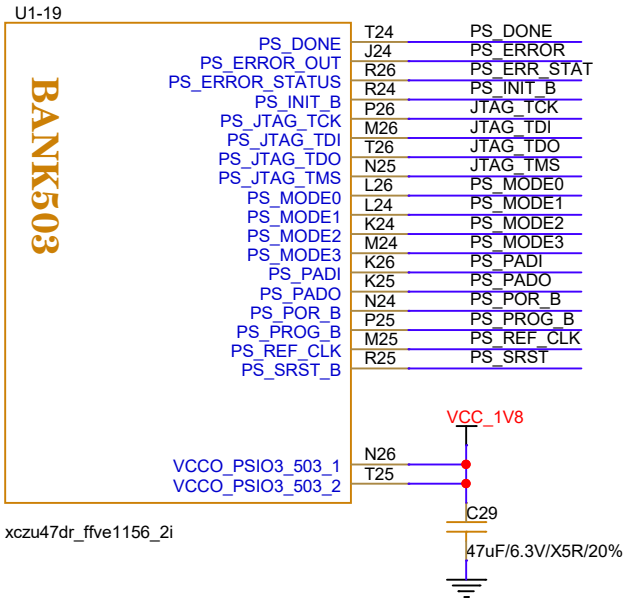
V3BEST

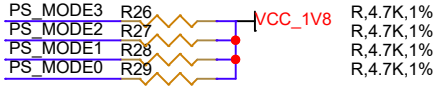
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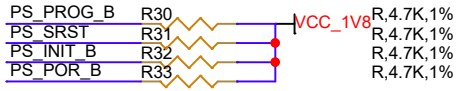
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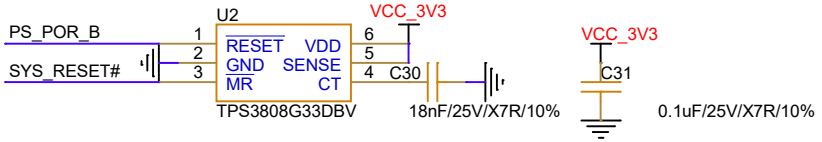
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U1-16
BANK500
PS_MIO0
PS_MIO1
PS_MIO2
PS_MIO3
PS_MIO4
PS_MIO5
PS_MIO6
PS_MIO7
PS_MIO8
PS_MIO9
PS_MIO10
PS_MIO11
PS_MIO12
PS_MIO13
PS_MIO14
PS_MIO15
PS_MIO16
PS_MIO17
PS_MIO18
PS_MIO19
PS_MIO20
PS_MIO21
PS_MIO22
PS_MIO23
PS_MIO24
PS_MIO25
J17 SPIL_SCLK
J18 SPIL_MO1
J16 SPIL_MO2
K16 SPIL_MO3
G15 SPIL_MO0
H18 SPIL_CS
H16
K17 SPIH_CS
E15 SPIH_MO0
F15 SPIH_MO1
C15 SPIH_MO2
G16 SPIH_MO3
B15 SPIH_SCLK
G18 LMK_REF_SEL0
A15 LMK_REF_SEL1
D16
G17 LMK_RESET
E16
F18 LMK_SPI_DOUT
C16
B16
F17 LMK_SPI_CS
E17 LMK_SPI_CLK
D17 LMK_SPI_DIN
A17 I2C_SCL
B17 I2C_SDA
LMK_REF_SEL0
LMK_REF_SEL1
LMK_RESET
LMK_SPI_DOUT
LMK_SPI_CS
LMK_SPI_CLK
LMK_SPI_DIN
VCC_1V8
VCCO_PSIO0_500_1
VCCO_PSIO0_500_2
C17
F16
C26
47uF/6.3V/X5R/20%
xczu47dr_ffve1156_2i
U1-18
BANK502
PS_MIO52
PS_MIO53
PS_MIO54
PS_MIO55
PS_MIO56
PS_MIO57
PS_MIO58
PS_MIO59
PS_MIO60
PS_MIO61
PS_MIO62
PS_MIO63
PS_MIO64
PS_MIO65
PS_MIO66
PS_MIO67
PS_MIO68
PS_MIO69
PS_MIO70
PS_MIO71
PS_MIO72
PS_MIO73
PS_MIO74
PS_MIO75
PS_MIO76
PS_MIO77
G22 BANK502_PS_MIO52
F22 BANK502_PS_MIO53
H23 BANK502_PS_MIO54
D22 BANK502_PS_MIO55
G23 BANK502_PS_MIO56
F23 BANK502_PS_MIO57
B22 BANK502_PS_MIO58
D23 BANK502_PS_MIO59
A23 BANK502_PS_MIO60
E22 BANK502_PS_MIO61
B23 BANK502_PS_MIO62
C23 BANK502_PS_MIO63
D24 BANK502_PS_MIO64
C24 BANK502_PS_MIO65
F24 BANK502_PS_MIO66
F25 BANK502_PS_MIO67
E25 BANK502_PS_MIO68
E24 BANK502_PS_MIO69
B25 BANK502_PS_MIO70
A24 BANK502_PS_MIO71
C25 BANK502_PS_MIO72
A25 BANK502_PS_MIO73
C26 BANK502_PS_MIO74
B26 BANK502_PS_MIO75
E26 BANK502_PS_MIO76
D26 BANK502_PS_MIO77
VCCO_502
VCCO_PSIO2_502_1
VCCO_PSIO2_502_2
E23
G25
C27
47uF/6.3V/X5R/20%
xczu47dr_ffve1156_2i
U1-17
BANK501
PS_MIO26
PS_MIO27
PS_MIO28
PS_MIO29
PS_MIO30
PS_MIO31
PS_MIO32
PS_MIO33
PS_MIO34
PS_MIO35
PS_MIO36
PS_MIO37
PS_MIO38
PS_MIO39
PS_MIO40
PS_MIO41
PS_MIO42
PS_MIO43
PS_MIO44
PS_MIO45
PS_MIO46
PS_MIO47
PS_MIO48
PS_MIO49
PS_MIO50
PS_MIO51
K19 BANK501_PS_MIO26
H19 BANK501_PS_MIO27
J19 BANK501_PS_MIO28
H21 BANK501_PS_MIO29
H20 BANK501_PS_MIO30
G20 BANK501_PS_MIO31
F19 BANK501_PS_MIO32
G21 BANK501_PS_MIO33
D18 BANK501_PS_MIO34
F20 BANK501_PS_MIO35
C18 BANK501_PS_MIO36
E19 BANK501_PS_MIO37
B18 BANK501_PS_MIO38
D19 BANK501_PS_MIO39
A18 BANK501_PS_MIO40
C19 BANK501_PS_MIO41
E20 BANK501_PS_MIO42
A19 BANK501_PS_MIO43
C20 BANK501_PS_MIO44
B20 BANK501_PS_MIO45
A20 BANK501_PS_MIO46
D21 BANK501_PS_MIO47
C21 BANK501_PS_MIO48
E21 BANK501_PS_MIO49
A22 BANK501_PS_MIO50
B21 BANK501_PS_MIO51
VCCO_501
VCCO_PSIO1_501_1
VCCO_PSIO1_501_2
D20
G19
C28
47uF/6.3V/X5R/20%
xczu47dr_ffve1156_2i
SPIL_SCLK
SPIL_CS
SPIL_MO[3:0]
SPIH_SCLK
SPIH_CS
SPIH_MO[3:0]
I2C_SDA
I2C_SCL
BANK501_PS_MIO26
BANK501_PS_MIO27
BANK501_PS_MIO28
BANK501_PS_MIO29
BANK501_PS_MIO30
BANK501_PS_MIO31
BANK501_PS_MIO32
BANK501_PS_MIO33
BANK501_PS_MIO34
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BANK501_PS_MIO38
BANK501_PS_MIO39
BANK501_PS_MIO40
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BANK501_PS_MIO42
BANK501_PS_MIO43
BANK501_PS_MIO44
BANK501_PS_MIO45
BANK501_PS_MIO46
BANK501_PS_MIO47
BANK501_PS_MIO48
BANK501_PS_MIO49
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BANK502_PS_MIO62
BANK502_PS_MIO63
BANK502_PS_MIO64
BANK502_PS_MIO65
BANK502_PS_MIO66
BANK502_PS_MIO67
BANK502_PS_MIO68
BANK502_PS_MIO69
BANK502_PS_MIO70
BANK502_PS_MIO71
BANK502_PS_MIO72
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BANK502_PS_MIO74
BANK502_PS_MIO75
BANK502_PS_MIO76
BANK502_PS_MIO77
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P08_FPGA_BANK_500-502
Rev
V1.3
Date:
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10
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33









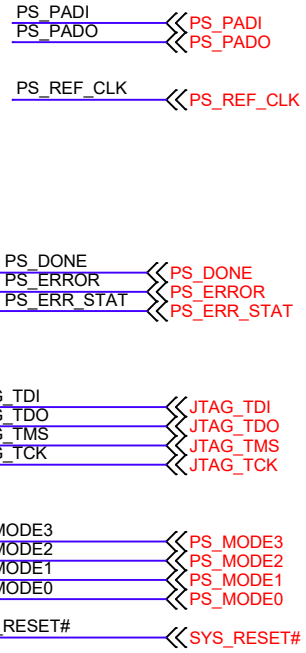
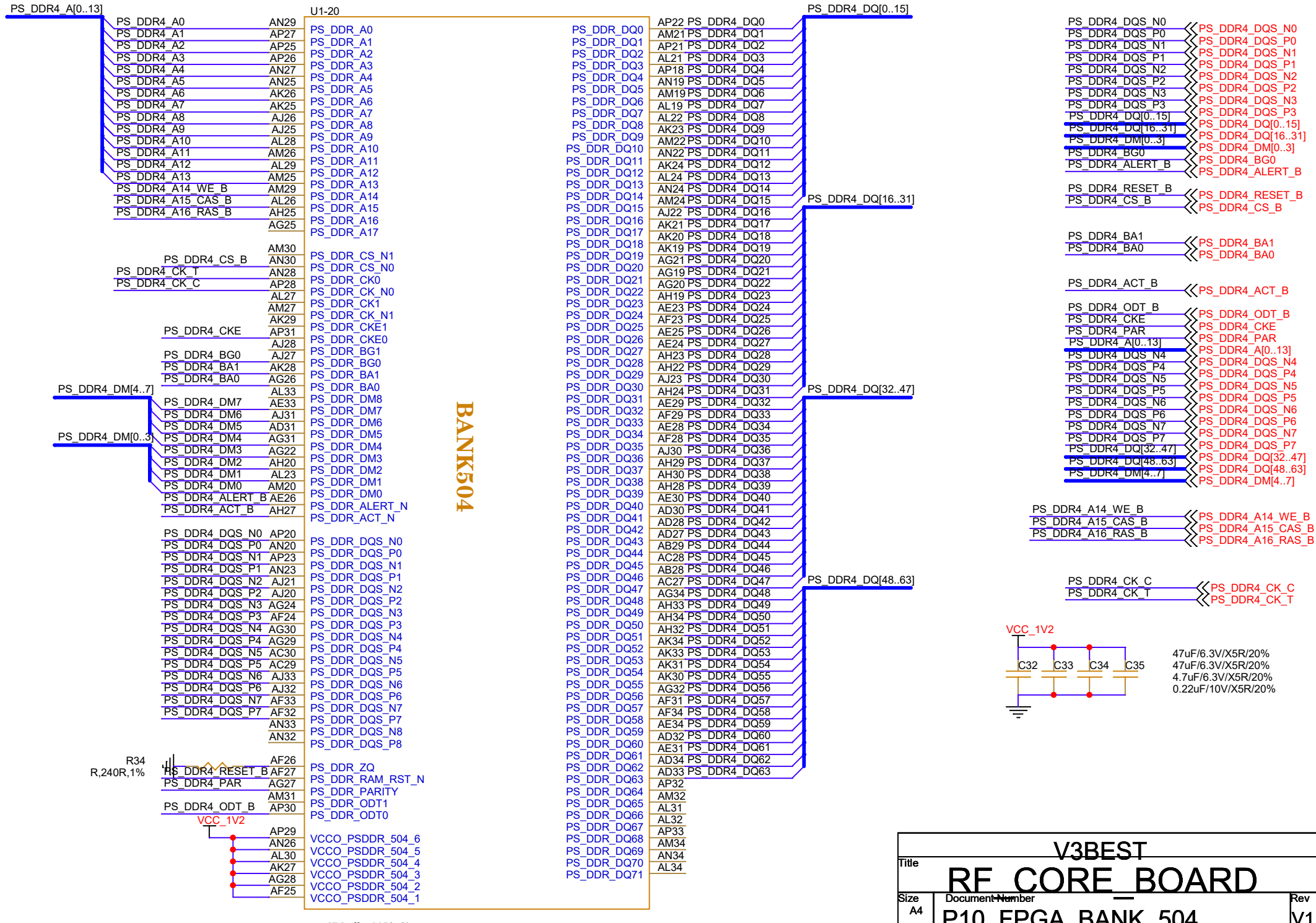


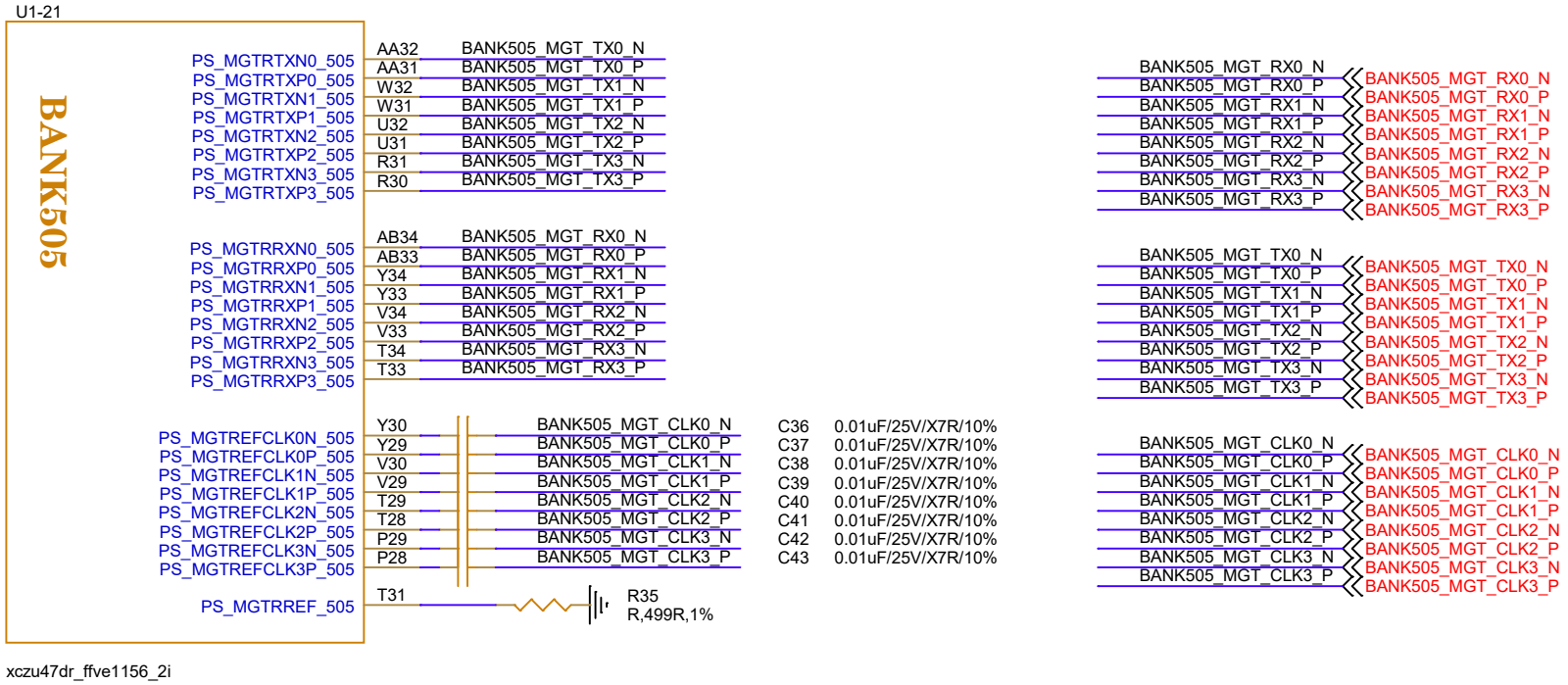
Table 11-1: **Boot Modes**

Boot Mode	Mode Pins [3:0]	Pin Location	CSU Mode	Description
PS JTAG	0000	JTAG	Slave	PSJTAG interface, PS dedicated pins.
Quad-SPI (24b)	0001	MIO[12:0]	Master	24-bit addressing (QSPI24).
Quad-SPI (32b)	0010	MIO[12:0]	Master	32-bit addressing (QSPI32).
SD0 (2.0)	0011	MIO[25:21, 16:13]	Master	SD 2.0.
NAND	0100	MIO[25:09]	Master	Requires 8-bit data bus width.
SD1 (2.0)	0101	MIO[51:43]	Master	SD 2.0.
eMMC (1.8V)	0110	MIO[22:13]	Master	eMMC version 4.5 at 1.8V.
USB0 (2.0)	0111	MIO[52:63]	Slave	USB 2.0 only.
PJTAG (MIO #0)	1000	MIO[29:26]	Slave	PJTAG connection 0 option.
PJTAG (MIO #1)	1001	MIO[15:12]	Slave	PJTAG connection 1 option.
SD1 LS (3.0)	1110	MIO[51:39]	Master	SD 3.0 with a required SD 3.0 compliant voltage level shifter.

```
PSMODE[3:0]
JTAG 0000
QSPI 0010
EMMC 0110
SD30 1110
```

Title	V3BEST RF_CORE_BOARD	
Size A4	Document Number P09_FPGA_BANK_503	Rev V1.3
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U1-20
BANK504
PS_DDR4_A[0..13]
PS_DDR4_A0 AN29
PS_DDR4_A1 AP27
PS_DDR4_A2 AP25
PS_DDR4_A3 AP26
PS_DDR4_A4 AN27
PS_DDR4_A5 AN25
PS_DDR4_A6 AK26
PS_DDR4_A7 AK25
PS_DDR4_A8 AJ26
PS_DDR4_A9 AJ25
PS_DDR4_A10 AL28
PS_DDR4_A11 AM26
PS_DDR4_A12 AL29
PS_DDR4_A13 AM25
PS_DDR4_A14_WE_B AM29
PS_DDR4_A15_CAS_B AL26
PS_DDR4_A16_RAS_B AH25
AG25
AM30
PS_DDR4_CS_B AN30
PS_DDR4_CK_T AN28
PS_DDR4_CK_C AP28
AL27
AM27
AK29
PS_DDR4_CKE AP31
AJ28
PS_DDR4_BG0 AJ27
PS_DDR4_BA1 AK28
PS_DDR4_BA0 AG26
PS_DDR4_DM[4..7]
AL33
PS_DDR4_DM7 AE33
PS_DDR4_DM6 AJ31
PS_DDR4_DM5 AD31
PS_DDR4_DM[0..3]
PS_DDR4_DM4 AG31
PS_DDR4_DM3 AG22
PS_DDR4_DM2 AH20
PS_DDR4_DM1 AL23
PS_DDR4_DM0 AM20
PS_DDR4_ALERT_B AE26
PS_DDR4_ACT_B AH27
PS_DDR4_DQS_N0 AP20
PS_DDR4_DQS_P0 AN20
PS_DDR4_DQS_N1 AP23
PS_DDR4_DQS_P1 AN23
PS_DDR4_DQS_N2 AJ21
PS_DDR4_DQS_P2 AJ20
PS_DDR4_DQS_N3 AG24
PS_DDR4_DQS_P3 AF24
PS_DDR4_DQS_N4 AG30
PS_DDR4_DQS_P4 AG29
PS_DDR4_DQS_N5 AC30
PS_DDR4_DQS_P5 AC29
PS_DDR4_DQS_N6 AJ33
PS_DDR4_DQS_P6 AJ32
PS_DDR4_DQS_N7 AF33
PS_DDR4_DQS_P7 AF32
AN33
AN32
R34
R,240R,1%
AF26
PS_DDR4_RESET_B AF27
PS_DDR4_PAR AG27
AM31
PS_DDR4_ODT_B AP30
VCC_1V2
AP29
AN26
AL30
AK27
AG28
AF25
PS_DDR_A0
PS_DDR_A1
PS_DDR_A2
PS_DDR_A3
PS_DDR_A4
PS_DDR_A5
PS_DDR_A6
PS_DDR_A7
PS_DDR_A8
PS_DDR_A9
PS_DDR_A10
PS_DDR_A11
PS_DDR_A12
PS_DDR_A13
PS_DDR_A14
PS_DDR_A15
PS_DDR_A16
PS_DDR_A17
PS_DDR_CS_N1
PS_DDR_CS_N0
PS_DDR_CK0
PS_DDR_CK_N0
PS_DDR_CK1
PS_DDR_CK_N1
PS_DDR_CKE1
PS_DDR_CKE0
PS_DDR_BG1
PS_DDR_BG0
PS_DDR_BA1
PS_DDR_BA0
PS_DDR_DM8
PS_DDR_DM7
PS_DDR_DM6
PS_DDR_DM5
PS_DDR_DM4
PS_DDR_DM3
PS_DDR_DM2
PS_DDR_DM1
PS_DDR_DM0
PS_DDR_ALERT_N
PS_DDR_ACT_N
PS_DDR_DQS_N0
PS_DDR_DQS_P0
PS_DDR_DQS_N1
PS_DDR_DQS_P1
PS_DDR_DQS_N2
PS_DDR_DQS_P2
PS_DDR_DQS_N3
PS_DDR_DQS_P3
PS_DDR_DQS_N4
PS_DDR_DQS_P4
PS_DDR_DQS_N5
PS_DDR_DQS_P5
PS_DDR_DQS_N6
PS_DDR_DQS_P6
PS_DDR_DQS_N7
PS_DDR_DQS_P7
PS_DDR_DQS_N8
PS_DDR_DQS_P8
PS_DDR_ZQ
PS_DDR_RAM_RST_N
PS_DDR_PARITY
PS_DDR_ODT1
PS_DDR_ODT0
VCCO_PSDDR_504_6
VCCO_PSDDR_504_5
VCCO_PSDDR_504_4
VCCO_PSDDR_504_3
VCCO_PSDDR_504_2
VCCO_PSDDR_504_1
xczu47dr_ffve1156_2i
PS_DDR_DQ0 AP22 PS_DDR4_DQ0
PS_DDR_DQ1 AM21 PS_DDR4_DQ1
PS_DDR_DQ2 AP21 PS_DDR4_DQ2
PS_DDR_DQ3 AL21 PS_DDR4_DQ3
PS_DDR_DQ4 AP18 PS_DDR4_DQ4
PS_DDR_DQ5 AN19 PS_DDR4_DQ5
PS_DDR_DQ6 AM19 PS_DDR4_DQ6
PS_DDR_DQ7 AL19 PS_DDR4_DQ7
PS_DDR_DQ8 AL22 PS_DDR4_DQ8
PS_DDR_DQ9 AK23 PS_DDR4_DQ9
PS_DDR_DQ10 AM22 PS_DDR4_DQ10
PS_DDR_DQ11 AN22 PS_DDR4_DQ11
PS_DDR_DQ12 AK24 PS_DDR4_DQ12
PS_DDR_DQ13 AL24 PS_DDR4_DQ13
PS_DDR_DQ14 AN24 PS_DDR4_DQ14
PS_DDR_DQ15 AM24 PS_DDR4_DQ15
PS_DDR_DQ16 AJ22 PS_DDR4_DQ16
PS_DDR_DQ17 AK21 PS_DDR4_DQ17
PS_DDR_DQ18 AK20 PS_DDR4_DQ18
PS_DDR_DQ19 AK19 PS_DDR4_DQ19
PS_DDR_DQ20 AG21 PS_DDR4_DQ20
PS_DDR_DQ21 AG19 PS_DDR4_DQ21
PS_DDR_DQ22 AG20 PS_DDR4_DQ22
PS_DDR_DQ23 AH19 PS_DDR4_DQ23
PS_DDR_DQ24 AE23 PS_DDR4_DQ24
PS_DDR_DQ25 AF23 PS_DDR4_DQ25
PS_DDR_DQ26 AE25 PS_DDR4_DQ26
PS_DDR_DQ27 AE24 PS_DDR4_DQ27
PS_DDR_DQ28 AH23 PS_DDR4_DQ28
PS_DDR_DQ29 AH22 PS_DDR4_DQ29
PS_DDR_DQ30 AJ23 PS_DDR4_DQ30
PS_DDR_DQ31 AH24 PS_DDR4_DQ31
PS_DDR_DQ32 AE29 PS_DDR4_DQ32
PS_DDR_DQ33 AF29 PS_DDR4_DQ33
PS_DDR_DQ34 AE28 PS_DDR4_DQ34
PS_DDR_DQ35 AF28 PS_DDR4_DQ35
PS_DDR_DQ36 AJ30 PS_DDR4_DQ36
PS_DDR_DQ37 AH29 PS_DDR4_DQ37
PS_DDR_DQ38 AH30 PS_DDR4_DQ38
PS_DDR_DQ39 AH28 PS_DDR4_DQ39
PS_DDR_DQ40 AE30 PS_DDR4_DQ40
PS_DDR_DQ41 AD30 PS_DDR4_DQ41
PS_DDR_DQ42 AD28 PS_DDR4_DQ42
PS_DDR_DQ43 AD27 PS_DDR4_DQ43
PS_DDR_DQ44 AB29 PS_DDR4_DQ44
PS_DDR_DQ45 AC28 PS_DDR4_DQ45
PS_DDR_DQ46 AB28 PS_DDR4_DQ46
PS_DDR_DQ47 AC27 PS_DDR4_DQ47
PS_DDR_DQ48 AG34 PS_DDR4_DQ48
PS_DDR_DQ49 AH33 PS_DDR4_DQ49
PS_DDR_DQ50 AH34 PS_DDR4_DQ50
PS_DDR_DQ51 AH32 PS_DDR4_DQ51
PS_DDR_DQ52 AK34 PS_DDR4_DQ52
PS_DDR_DQ53 AK33 PS_DDR4_DQ53
PS_DDR_DQ54 AK31 PS_DDR4_DQ54
PS_DDR_DQ55 AK30 PS_DDR4_DQ55
PS_DDR_DQ56 AG32 PS_DDR4_DQ56
PS_DDR_DQ57 AF31 PS_DDR4_DQ57
PS_DDR_DQ58 AF34 PS_DDR4_DQ58
PS_DDR_DQ59 AE34 PS_DDR4_DQ59
PS_DDR_DQ60 AD32 PS_DDR4_DQ60
PS_DDR_DQ61 AE31 PS_DDR4_DQ61
PS_DDR_DQ62 AD34 PS_DDR4_DQ62
PS_DDR_DQ63 AD33 PS_DDR4_DQ63
AP32
AM32
AL31
AL32
AP33
AM34
AN34
AL34
PS_DDR4_DQ[0..15]
PS_DDR4_DQ[16..31]
PS_DDR4_DQ[32..47]
PS_DDR4_DQ[48..63]
PS_DDR4_DQS_N0
PS_DDR4_DQS_P0
PS_DDR4_DQS_N1
PS_DDR4_DQS_P1
PS_DDR4_DQS_N2
PS_DDR4_DQS_P2
PS_DDR4_DQS_N3
PS_DDR4_DQS_P3
PS_DDR4_DQ[0..15]
PS_DDR4_DQ[16..31]
PS_DDR4_DM[0..3]
PS_DDR4_BG0
PS_DDR4_ALERT_B
PS_DDR4_RESET_B
PS_DDR4_CS_B
PS_DDR4_BA1
PS_DDR4_BA0
PS_DDR4_ACT_B
PS_DDR4_ODT_B
PS_DDR4_CKE
PS_DDR4_PAR
PS_DDR4_A[0..13]
PS_DDR4_DQS_N4
PS_DDR4_DQS_P4
PS_DDR4_DQS_N5
PS_DDR4_DQS_P5
PS_DDR4_DQS_N6
PS_DDR4_DQS_P6
PS_DDR4_DQS_N7
PS_DDR4_DQS_P7
PS_DDR4_DQ[32..47]
PS_DDR4_DQ[48..63]
PS_DDR4_DM[4..7]
PS_DDR4_A14_WE_B
PS_DDR4_A15_CAS_B
PS_DDR4_A16_RAS_B
PS_DDR4_CK_C
PS_DDR4_CK_T
VCC_1V2
C32
C33
C34
C35
47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P10_FPGA_BANK_504
Rev
V1.3
Date: Thursday, November 03, 2022
Sheet 12 of 33



Title	V3BEST RF_CORE_BOARD		
Size A4	Document Number P11_FPGA_BANK_505		Rev V1.3
Date: Thursday, November 03, 2022	Sheet 13 of 33		

AVCC_DAC_0V925

0.925V 3A

U1-26

DAC_AVCC_1 G6
DAC_AVCC_2 G7
DAC_AVCC_3 H7
DAC_AVCC_4 J7
DAC_AVCC_5 K7
DAC_AVCC_6 L7
DAC_AVCC_7 M7
DAC_AVCC_8 N7
DAC_AVCC_9 P7

xczu47dr_ffve1156_2i

AUX_DAC_1V8

1.8V 0.1A

U1-27

DAC_AVCCAUX_1 D6
DAC_AVCCAUX_2 D7
DAC_AVCCAUX_3 E6
DAC_AVCCAUX_4 E7

xczu47dr_ffve1156_2i

AVTT_DAC_2V5

2.5V 0.5A

U1-28

DAC_AVTT_1 D4
DAC_AVTT_2 E4
DAC_AVTT_3 F4
DAC_AVTT_4 G4

xczu47dr_ffve1156_2i

U1-30

DAC_SUB_GND L8

xczu47dr_ffve1156_2i

AVCC_ADC_0V925

0.925V 3A

U1-22

ADC_AVCC_1 AA7
ADC_AVCC_2 AB7
ADC_AVCC_3 AC7
ADC_AVCC_4 AD7
ADC_AVCC_5 AE7
ADC_AVCC_6 W7
ADC_AVCC_7 Y7

xczu47dr_ffve1156_2i

AUX_ADC_1V8

1.8V 3A

U1-23

ADC_AVCCAUX_1 AG7
ADC_AVCCAUX_2 AH7
ADC_AVCCAUX_3 AJ7
ADC_AVCCAUX_4 AK7

xczu47dr_ffve1156_2i

U1-25

ADC_SUB_GND AC8

xczu47dr_ffve1156_2i

AVCC_DAC_0V925

C44 C45 C46 C47

47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%

AVCC_DAC_0V925

C48 C49 C50 C51 C52

4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
0.22uF/10V/X5R/20%

AUX_DAC_1V8

C53 C54 C55 C56

47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%

AUX_DAC_1V8

C57 C58 C59 C60 C61

4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
0.22uF/10V/X5R/20%

AVTT_DAC_2V5

C62 C63 C64

47uF/6.3V/X5R/20%

4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%

AVCC_ADC_0V925

C65 C66 C67 C68

47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%

AVCC_ADC_0V925

C69 C70 C71 C72 C73

4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
0.22uF/10V/X5R/20%

AUX_ADC_1V8

C74 C75 C76 C77

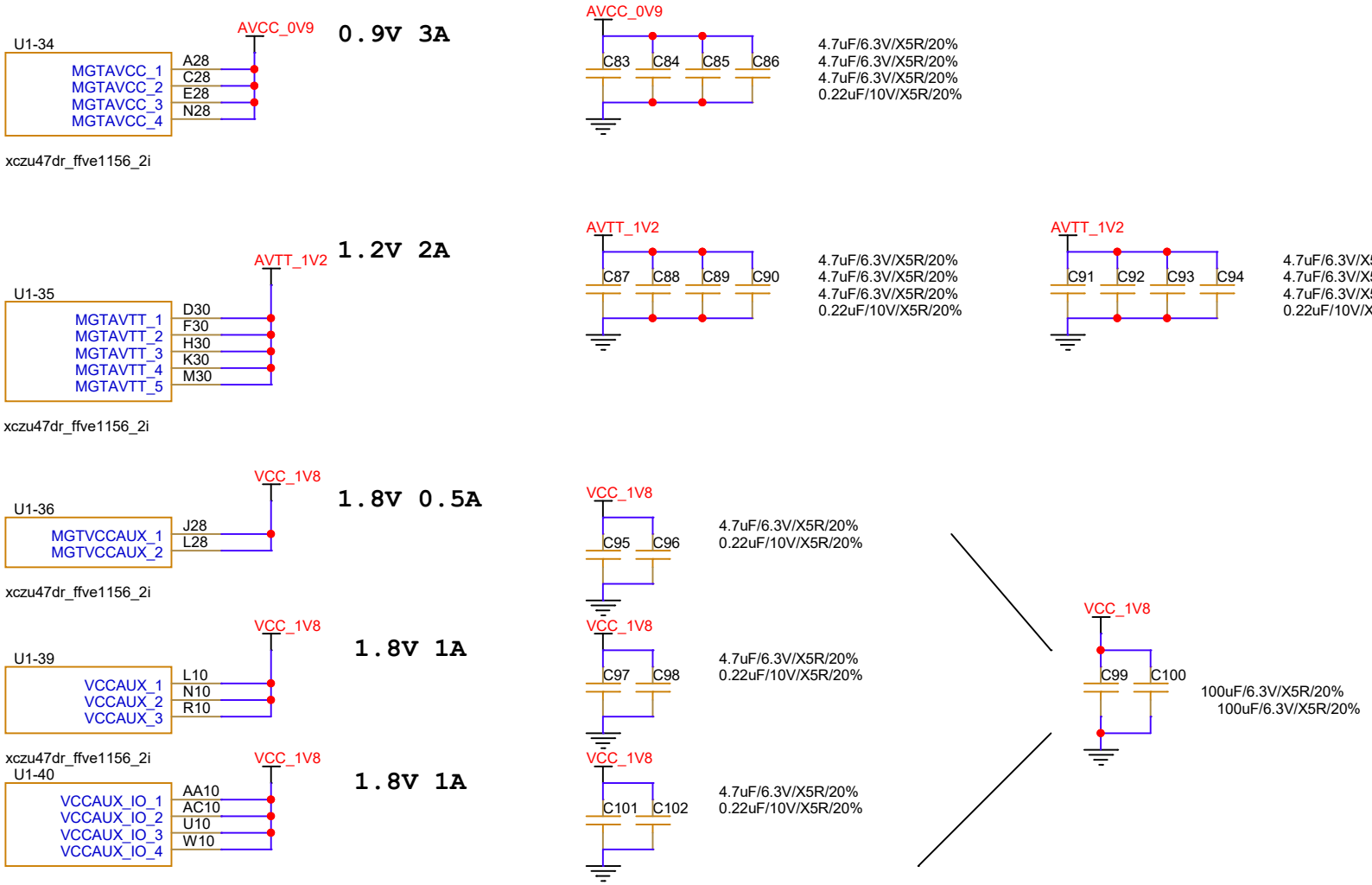
47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%

AUX_ADC_1V8

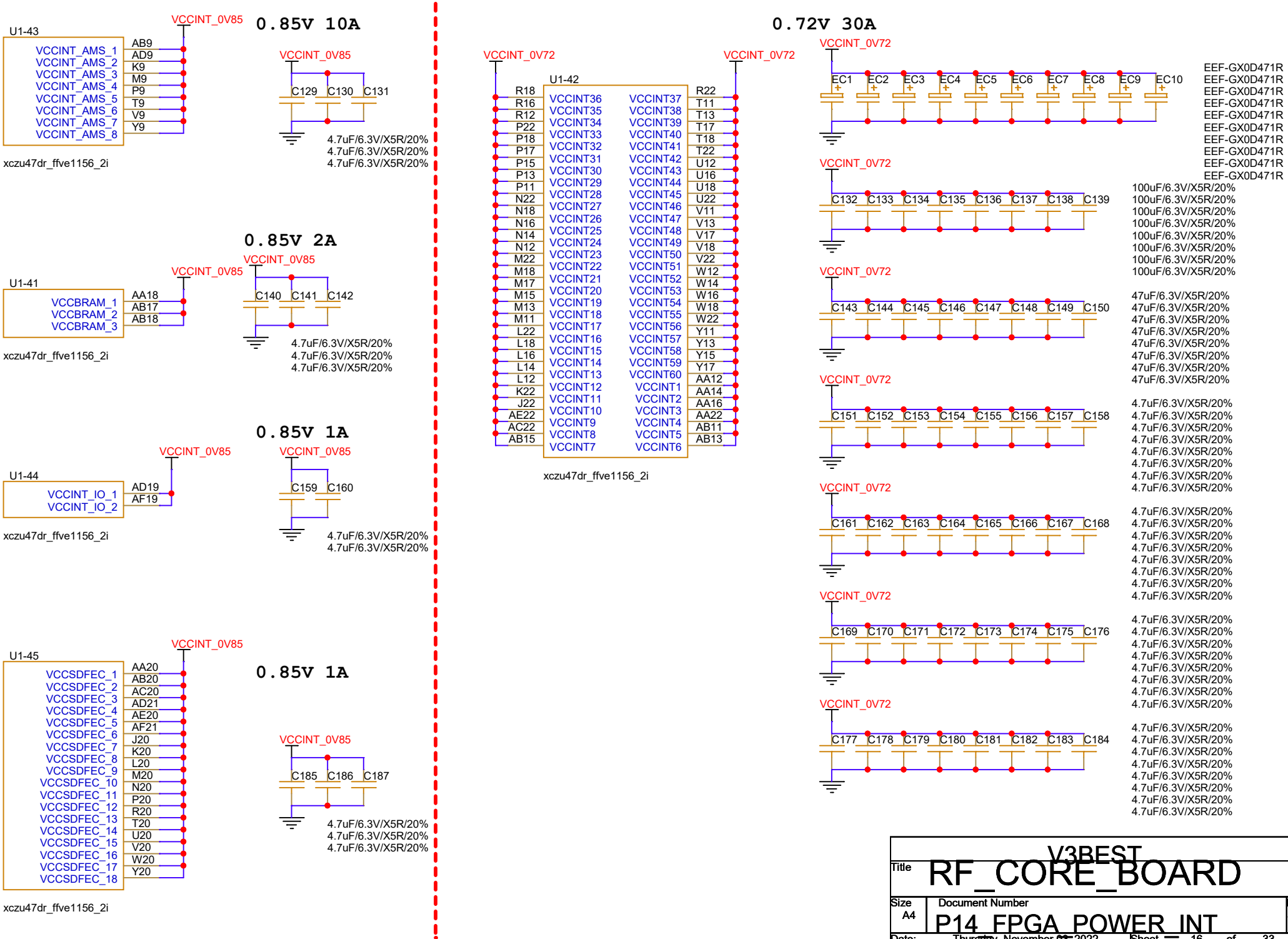
C78 C79 C80 C81 C82

4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
0.22uF/10V/X5R/20%

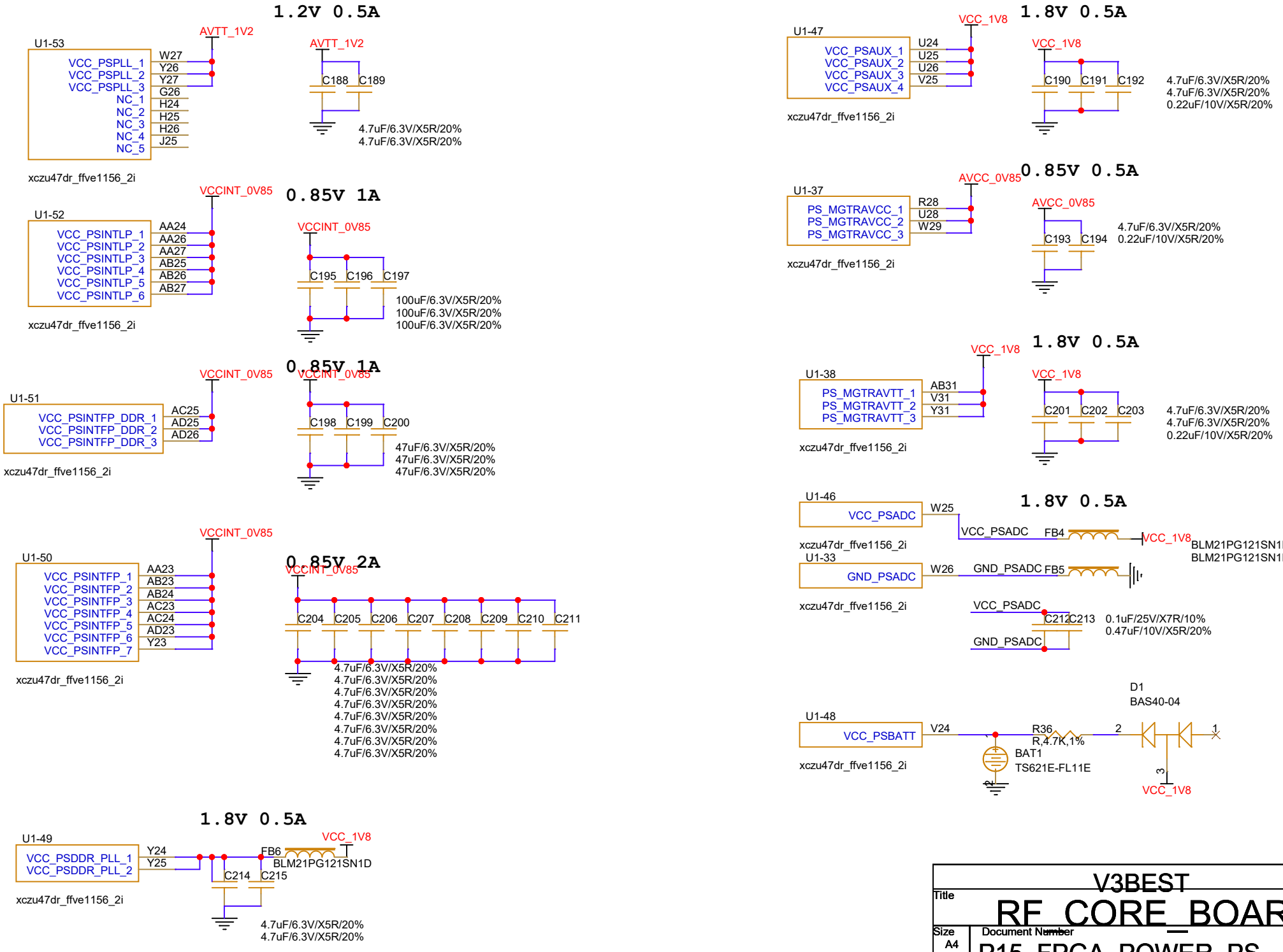
Title	V3BEST RF_CORE_BOARD	
Size A4	Document Number P12_FPGA_ANALOG_POWER	Rev V1.3
Date: Thursday, November 03, 2022	Sheet 14 of 33	

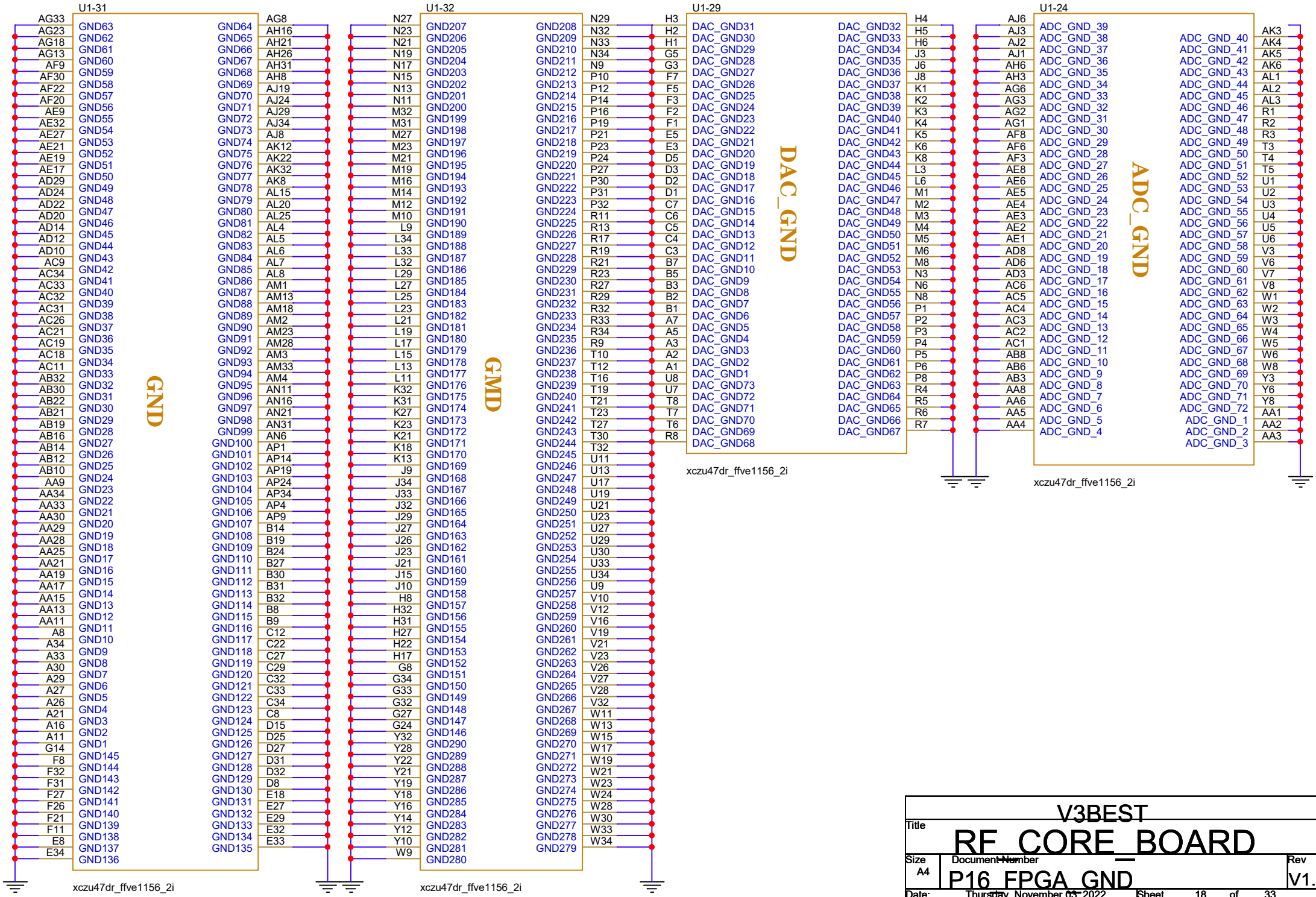


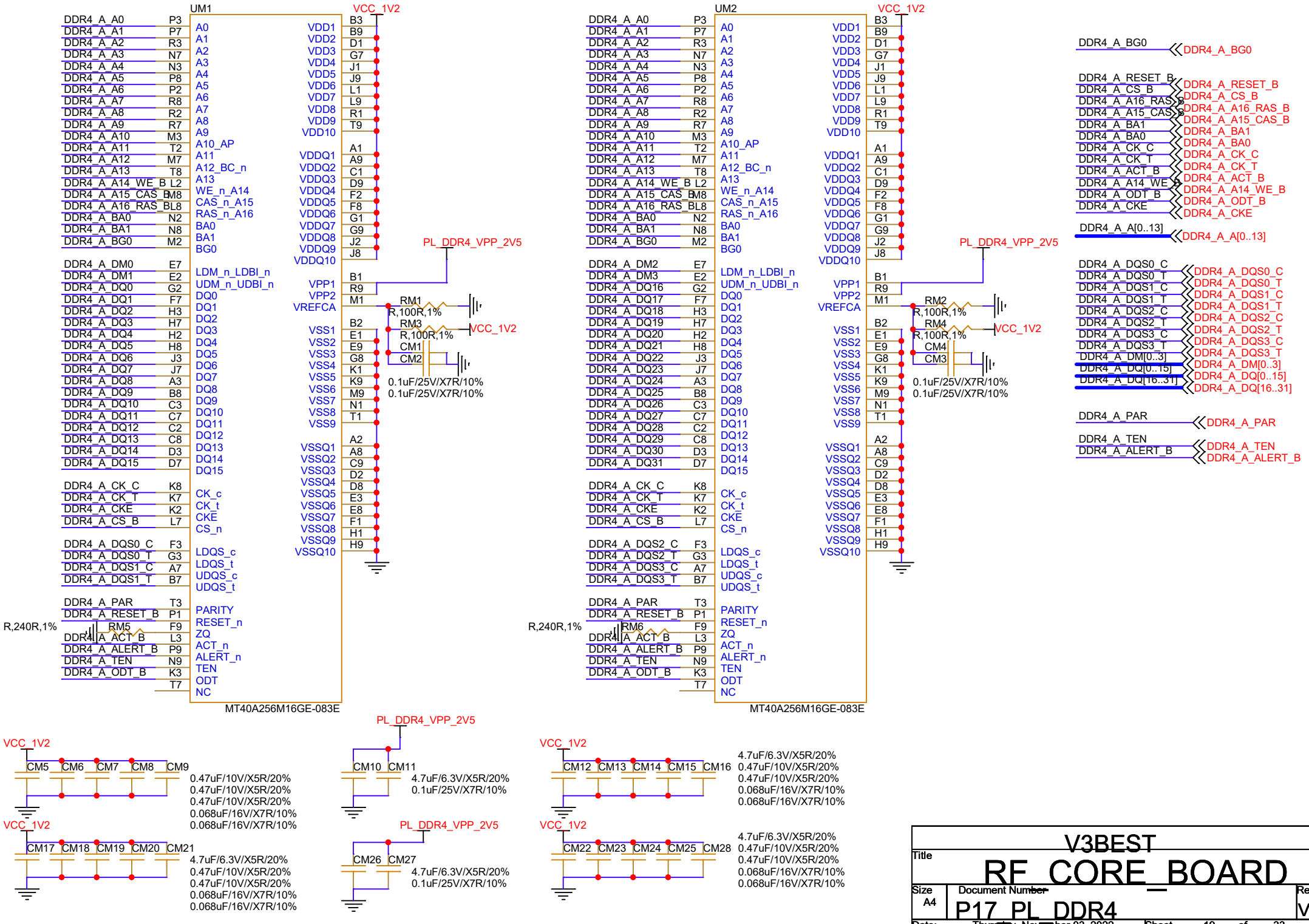
V3BEST			
Title	RF_CORE_BOARD		
Size A4	Document Number P13_FPGA_POWER_GTY_AUX		Rev V1.3
Date: Thursday, November 03, 2022	Sheet 15 of 33		

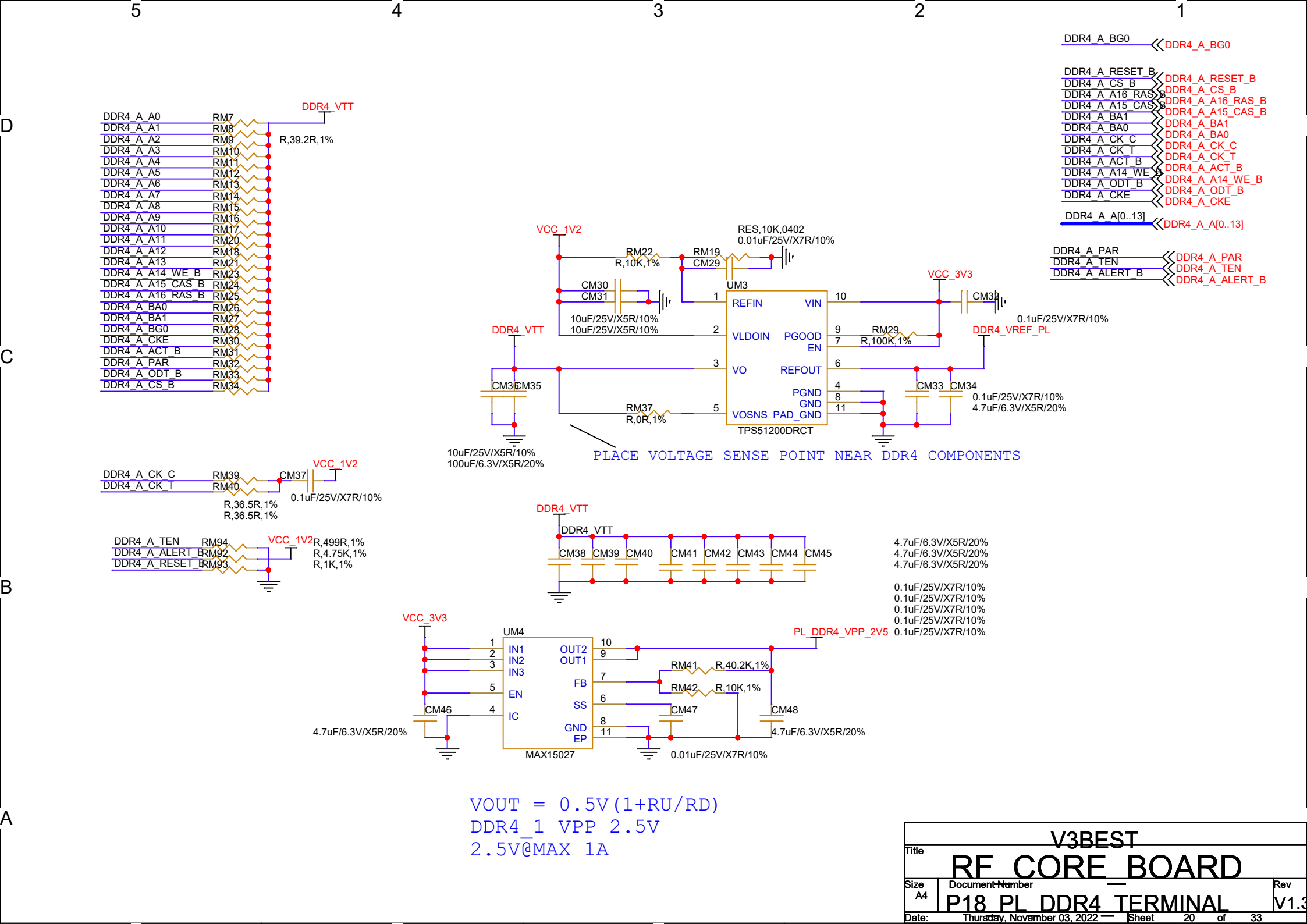


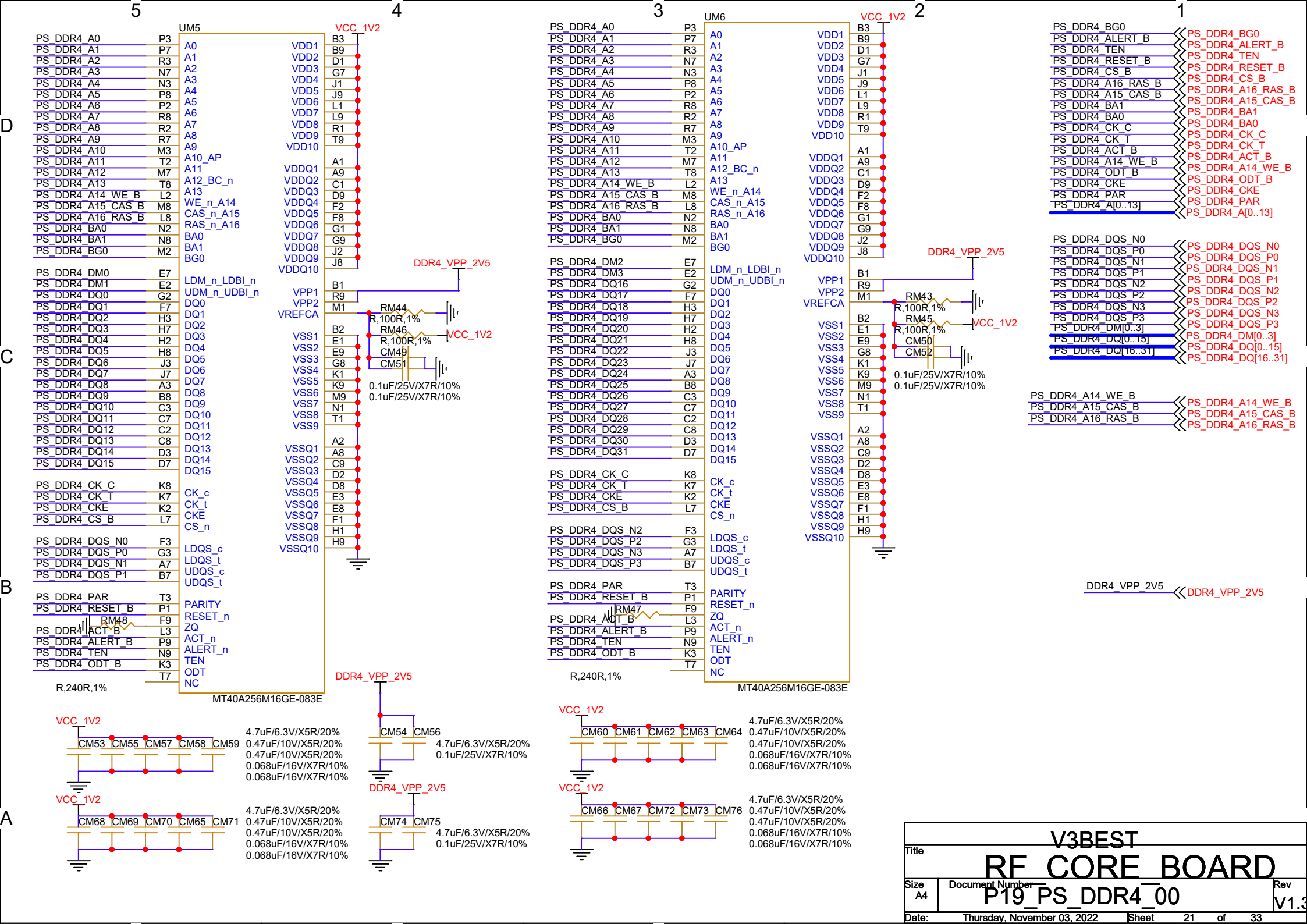
	V3BEST	
Title	RF_CORE_BOARD	
Size A4	Document Number P14_FPGA_POWER_INT	Rev V1.3
Date: Thursday, November 03, 2022	Sheet — 16 of 33	

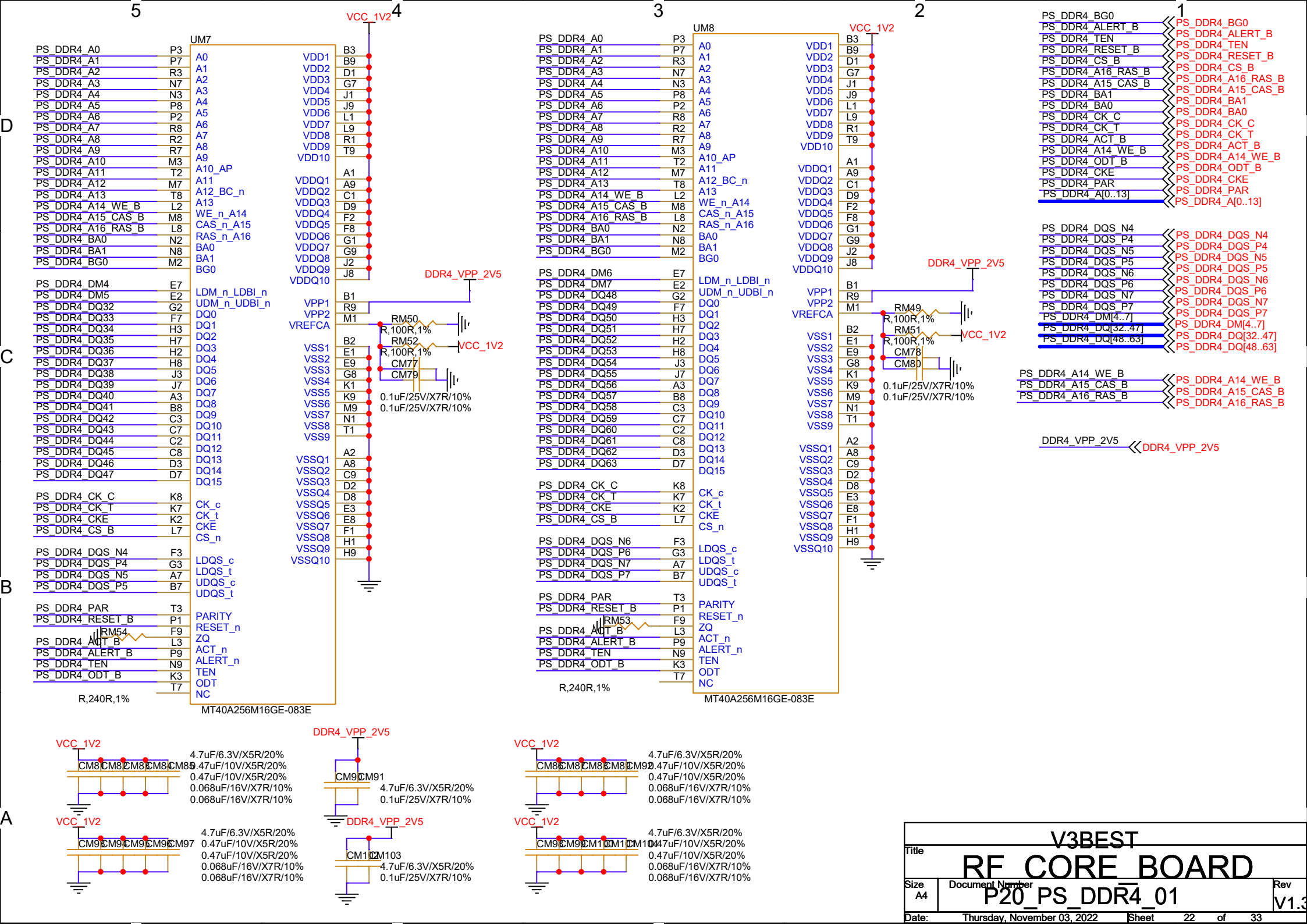

1.2V 0.5A
U1-53
VCC_PSPLL_1
VCC_PSPLL_2
VCC_PSPLL_3
NC_1
NC_2
NC_3
NC_4
NC_5
W27
Y26
Y27
G26
H24
H25
H26
J25
AVTT_1V2
xczu47dr_ffve1156_2i
C188
C189
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.85V 1A
U1-52
VCC_PSINTLP_1
VCC_PSINTLP_2
VCC_PSINTLP_3
VCC_PSINTLP_4
VCC_PSINTLP_5
VCC_PSINTLP_6
AA24
AA26
AA27
AB25
AB26
AB27
VCCINT_0V85
C195
C196
C197
100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%
0.85V 1A
U1-51
VCC_PSINTFP_DDR_1
VCC_PSINTFP_DDR_2
VCC_PSINTFP_DDR_3
AC25
AD25
AD26
C198
C199
C200
47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
47uF/6.3V/X5R/20%
0.85V 2A
U1-50
VCC_PSINTFP_1
VCC_PSINTFP_2
VCC_PSINTFP_3
VCC_PSINTFP_4
VCC_PSINTFP_5
VCC_PSINTFP_6
VCC_PSINTFP_7
AA23
AB23
AB24
AC23
AC24
AD23
Y23
C204
C205
C206
C207
C208
C209
C210
C211
4.7uF/6.3V/X5R/20%
1.8V 0.5A
U1-49
VCC_PSDDR_PLL_1
VCC_PSDDR_PLL_2
Y24
Y25
FB6
BLM21PG121SN1D
C214
C215
4.7uF/6.3V/X5R/20%
1.8V 0.5A
U1-47
VCC_PSAUX_1
VCC_PSAUX_2
VCC_PSAUX_3
VCC_PSAUX_4
U24
U25
U26
V25
C190
C191
C192
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
0.85V 0.5A
U1-37
PS_MGTRAVCC_1
PS_MGTRAVCC_2
PS_MGTRAVCC_3
R28
U28
W29
AVCC_0V85
C193
C194
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
1.8V 0.5A
U1-38
PS_MGTRAVTT_1
PS_MGTRAVTT_2
PS_MGTRAVTT_3
AB31
V31
Y31
C201
C202
C203
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.22uF/10V/X5R/20%
1.8V 0.5A
U1-46
VCC_PSADC
W25
FB4
BLM21PG121SN1D
U1-33
GND_PSADC
W26
FB5
BLM21PG121SN1D
C212
C213
0.1uF/25V/X7R/10%
0.47uF/10V/X5R/20%
D1
BAS40-04
U1-48
VCC_PSBATT
V24
R36
R,4.7K,1%
BAT1
TS621E-FL11E
V3BEST
RF_CORE_BOARD
P15_FPGA_POWER_PS
V1.3
Thursday, November 03, 2022
17
33

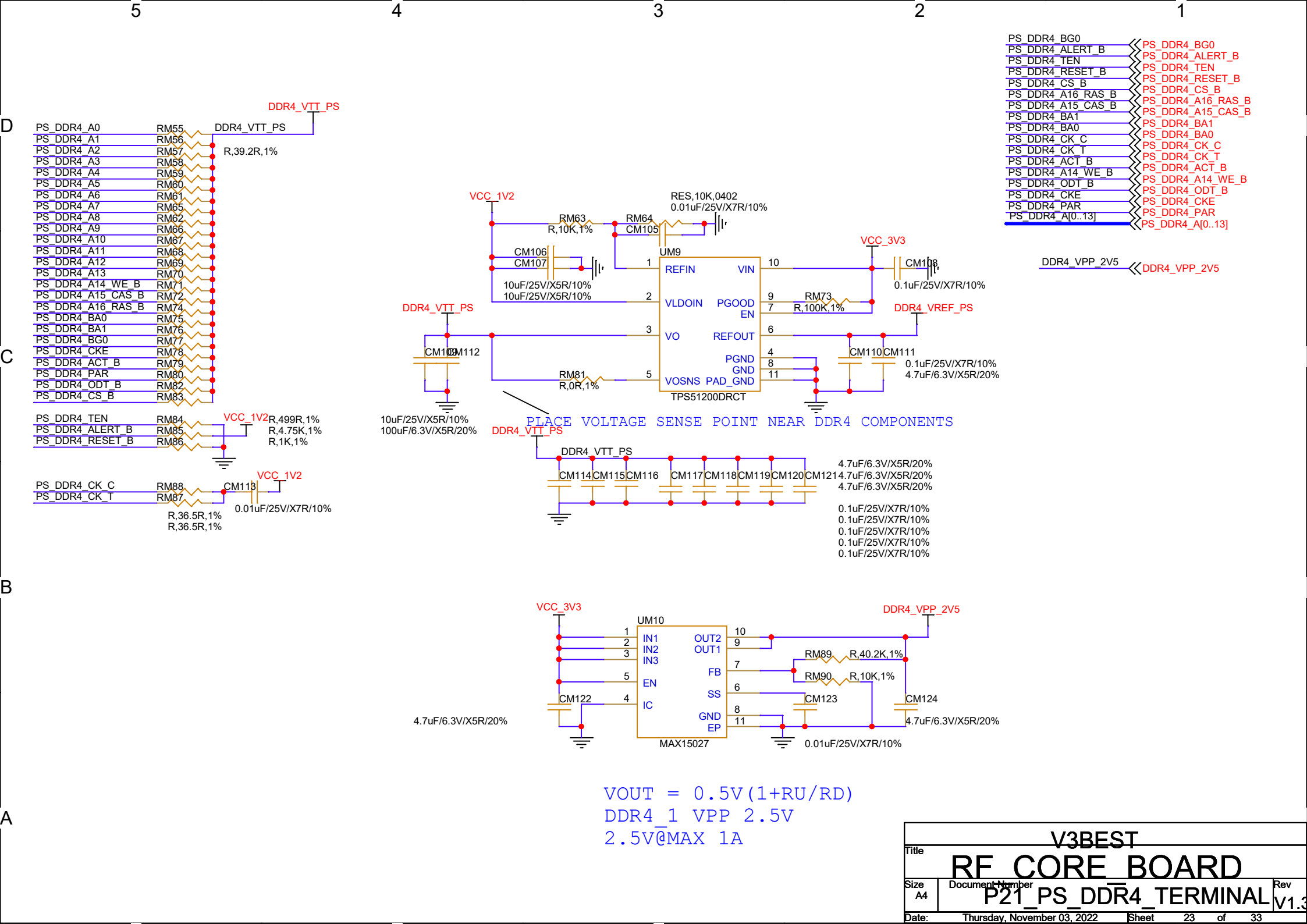

U1-31
GND
xczu47dr_ffve1156_2i
U1-32
GMD
xczu47dr_ffve1156_2i
U1-29
DAC_GND
xczu47dr_ffve1156_2i
U1-24
ADC_GND
xczu47dr_ffve1156_2i
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P16_FPGA_GND
Rev
V1.3
Date: Thursday, November 03, 2022
Sheet 18 of 33

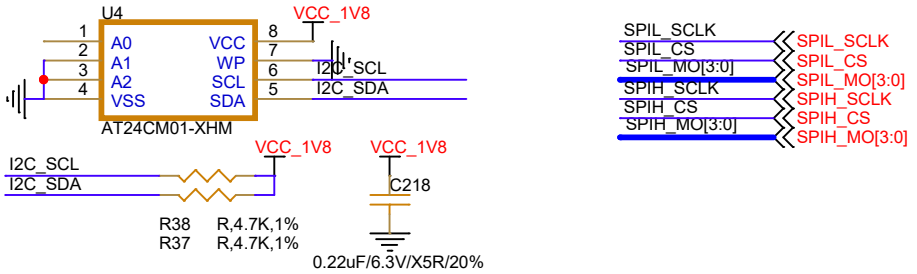

UM1
MT40A256M16GE-083E
UM2
MT40A256M16GE-083E
VCC_1V2
PL_DDR4_VPP_2V5
RM1
RM2
RM3
RM4
RM5
RM6
R,100R,1%
R,240R,1%
CM1
CM2
CM3
CM4
0.1uF/25V/X7R/10%
DDR4_A_BG0
DDR4_A_RESET_B
DDR4_A_CS_B
DDR4_A_A16_RAS_B
DDR4_A_A15_CAS_B
DDR4_A_BA1
DDR4_A_BA0
DDR4_A_CK_C
DDR4_A_CK_T
DDR4_A_ACT_B
DDR4_A_A14_WE_B
DDR4_A_ODT_B
DDR4_A_CKE
DDR4_A_A[0..13]
DDR4_A_DQS0_C
DDR4_A_DQS0_T
DDR4_A_DQS1_C
DDR4_A_DQS1_T
DDR4_A_DQS2_C
DDR4_A_DQS2_T
DDR4_A_DQS3_C
DDR4_A_DQS3_T
DDR4_A_DM[0..3]
DDR4_A_DQ[0..15]
DDR4_A_DQ[16..31]
DDR4_A_PAR
DDR4_A_TEN
DDR4_A_ALERT_B
CM5 CM6 CM7 CM8 CM9
0.47uF/10V/X5R/20%
0.47uF/10V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
0.068uF/16V/X7R/10%
CM17 CM18 CM19 CM20 CM21
4.7uF/6.3V/X5R/20%
0.47uF/10V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
0.068uF/16V/X7R/10%
CM10 CM11
4.7uF/6.3V/X5R/20%
0.1uF/25V/X7R/10%
CM26 CM27
4.7uF/6.3V/X5R/20%
0.1uF/25V/X7R/10%
CM12 CM13 CM14 CM15 CM16
4.7uF/6.3V/X5R/20%
0.47uF/10V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
0.068uF/16V/X7R/10%
CM22 CM23 CM24 CM25 CM28
4.7uF/6.3V/X5R/20%
0.47uF/10V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
0.068uF/16V/X7R/10%
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P17_PL_DDR4
Rev
V1.3
Date:
Thursday, November 03, 2022
Sheet
19
of
33

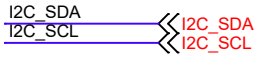

DDR4_A_BG0
DDR4_A_RESET_B
DDR4_A_CS_B
DDR4_A_A16_RAS_B
DDR4_A_A15_CAS_B
DDR4_A_BA1
DDR4_A_BA0
DDR4_A_CK_C
DDR4_A_CK_T
DDR4_A_ACT_B
DDR4_A_A14_WE_B
DDR4_A_ODT_B
DDR4_A_CKE
DDR4_A_A[0..13]
DDR4_A_PAR
DDR4_A_TEN
DDR4_A_ALERT_B
DDR4_VTT
R,39.2R,1%
RM7
RM8
RM9
RM10
RM11
RM12
RM13
RM14
RM15
RM16
RM17
RM20
RM18
RM21
RM23
RM24
RM25
RM26
RM27
RM28
RM30
RM31
RM32
RM33
RM34
VCC_1V2
RES,10K,0402
0.01uF/25V/X7R/10%
RM22
R,10K,1%
RM19
CM29
CM30
CM31
10uF/25V/X5R/10%
10uF/25V/X5R/10%
UM3
REFIN
VLDOIN
VO
VOSNS
VIN
PGOOD
EN
REFOUT
PGND
GND
PAD_GND
TPS51200DRCT
VCC_3V3
CM32
0.1uF/25V/X7R/10%
RM29
R,100K,1%
DDR4_VREF_PL
CM33
CM34
0.1uF/25V/X7R/10%
4.7uF/6.3V/X5R/20%
RM37
R,0R,1%
CM35
10uF/25V/X5R/10%
100uF/6.3V/X5R/20%
PLACE VOLTAGE SENSE POINT NEAR DDR4 COMPONENTS
DDR4_A_CK_C
DDR4_A_CK_T
RM39
RM40
CM37
0.1uF/25V/X7R/10%
R,36.5R,1%
R,36.5R,1%
DDR4_A_TEN
DDR4_A_ALERT_B
DDR4_A_RESET_B
RM94
RM92
RM93
R,499R,1%
R,4.75K,1%
R,1K,1%
CM38
CM39
CM40
CM41
CM42
CM43
CM44
CM45
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
UM4
IN1
IN2
IN3
EN
IC
OUT2
OUT1
FB
SS
GND
EP
MAX15027
PL_DDR4_VPP_2V5
RM41
R,40.2K,1%
RM42
R,10K,1%
CM46
CM47
CM48
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.01uF/25V/X7R/10%
VOUT = 0.5V(1+RU/RD)
DDR4_1 VPP 2.5V
2.5V@MAX 1A
V3BEST
RF_CORE_BOARD
P18_PL_DDR4_TERMINAL
A4
V1.3
Thursday, November 03, 2022
20
33

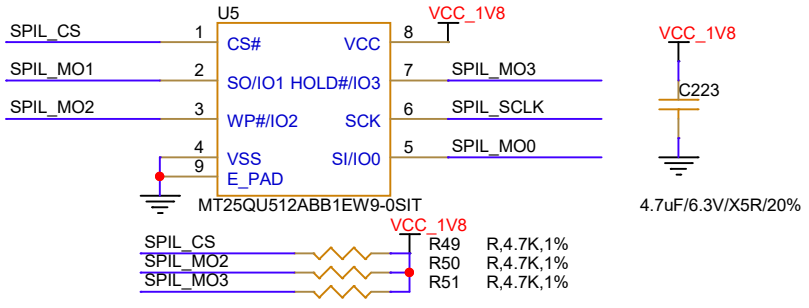

5
4
3
2
1
D
C
B
A
UM5
MT40A256M16GE-083E
UM6
MT40A256M16GE-083E
VCC_1V2
DDR4_VPP_2V5
R,240R,1%
R,100R,1%
RM44
RM46
RM43
RM45
RM47
RM48
CM49
CM51
CM50
CM52
0.1uF/25V/X7R/10%
CM53 CM55 CM57 CM58 CM59
CM54 CM56
CM60 CM61 CM62 CM63 CM64
CM68 CM69 CM70 CM65 CM71
CM74 CM75
CM66 CM67 CM72 CM73 CM76
4.7uF/6.3V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
0.1uF/25V/X7R/10%
PS_DDR4_BG0
PS_DDR4_ALERT_B
PS_DDR4_TEN
PS_DDR4_RESET_B
PS_DDR4_CS_B
PS_DDR4_A16_RAS_B
PS_DDR4_A15_CAS_B
PS_DDR4_BA1
PS_DDR4_BA0
PS_DDR4_CK_C
PS_DDR4_CK_T
PS_DDR4_ACT_B
PS_DDR4_A14_WE_B
PS_DDR4_ODT_B
PS_DDR4_CKE
PS_DDR4_PAR
PS_DDR4_A[0..13]
PS_DDR4_DQS_N0
PS_DDR4_DQS_P0
PS_DDR4_DQS_N1
PS_DDR4_DQS_P1
PS_DDR4_DQS_N2
PS_DDR4_DQS_P2
PS_DDR4_DQS_N3
PS_DDR4_DQS_P3
PS_DDR4_DM[0..3]
PS_DDR4_DQ[0..15]
PS_DDR4_DQ[16..31]
V3BEST
Title
RF_CORE_BOARD
Size
A4
Document Number
P19_PS_DDR4_00
Rev
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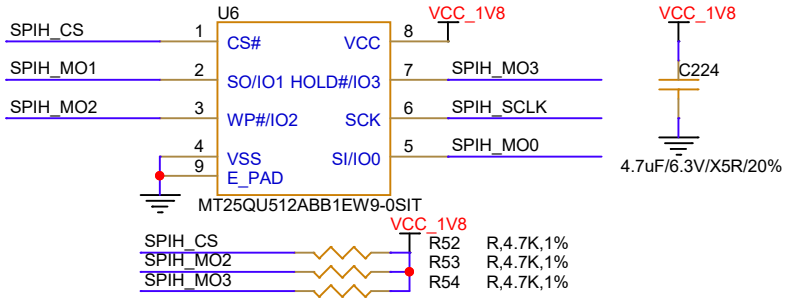

5
4
3
2
1
UM7
UM8
MT40A256M16GE-083E
VCC_1V2
DDR4_VPP_2V5
PS_DDR4_A0
PS_DDR4_A1
PS_DDR4_A2
PS_DDR4_A3
PS_DDR4_A4
PS_DDR4_A5
PS_DDR4_A6
PS_DDR4_A7
PS_DDR4_A8
PS_DDR4_A9
PS_DDR4_A10
PS_DDR4_A11
PS_DDR4_A12
PS_DDR4_A13
PS_DDR4_A14_WE_B
PS_DDR4_A15_CAS_B
PS_DDR4_A16_RAS_B
PS_DDR4_BA0
PS_DDR4_BA1
PS_DDR4_BG0
PS_DDR4_DM4
PS_DDR4_DM5
PS_DDR4_DQ32
PS_DDR4_DQ33
PS_DDR4_DQ34
PS_DDR4_DQ35
PS_DDR4_DQ36
PS_DDR4_DQ37
PS_DDR4_DQ38
PS_DDR4_DQ39
PS_DDR4_DQ40
PS_DDR4_DQ41
PS_DDR4_DQ42
PS_DDR4_DQ43
PS_DDR4_DQ44
PS_DDR4_DQ45
PS_DDR4_DQ46
PS_DDR4_DQ47
PS_DDR4_CK_C
PS_DDR4_CK_T
PS_DDR4_CKE
PS_DDR4_CS_B
PS_DDR4_DQS_N4
PS_DDR4_DQS_P4
PS_DDR4_DQS_N5
PS_DDR4_DQS_P5
PS_DDR4_PAR
PS_DDR4_RESET_B
PS_DDR4_ACT_B
PS_DDR4_ALERT_B
PS_DDR4_TEN
PS_DDR4_ODT_B
RM54
RM50
RM52
RM53
RM49
RM51
R,100R,1%
R,240R,1%
CM77
CM79
CM78
CM80
0.1uF/25V/X7R/10%
4.7uF/6.3V/X5R/20%
0.47uF/10V/X5R/20%
0.068uF/16V/X7R/10%
CM81
CM82
CM83
CM84
CM85
CM86
CM87
CM88
CM89
CM90
CM91
CM92
CM93
CM94
CM95
CM96
CM97
CM98
CM99
CM100
CM101
CM102
CM103
CM104
PS_DDR4_BG0
PS_DDR4_ALERT_B
PS_DDR4_TEN
PS_DDR4_RESET_B
PS_DDR4_CS_B
PS_DDR4_A16_RAS_B
PS_DDR4_A15_CAS_B
PS_DDR4_BA1
PS_DDR4_BA0
PS_DDR4_CK_C
PS_DDR4_CK_T
PS_DDR4_ACT_B
PS_DDR4_A14_WE_B
PS_DDR4_ODT_B
PS_DDR4_CKE
PS_DDR4_PAR
PS_DDR4_A[0..13]
PS_DDR4_DQS_N4
PS_DDR4_DQS_P4
PS_DDR4_DQS_N5
PS_DDR4_DQS_P5
PS_DDR4_DQS_N6
PS_DDR4_DQS_P6
PS_DDR4_DQS_N7
PS_DDR4_DQS_P7
PS_DDR4_DM[4..7]
PS_DDR4_DQ[32..47]
PS_DDR4_DQ[48..63]
PS_DDR4_A14_WE_B
PS_DDR4_A15_CAS_B
PS_DDR4_A16_RAS_B
DDR4_VPP_2V5
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5
4
3
2
1
D
C
B
A
PS_DDR4_BG0
PS_DDR4_ALERT_B
PS_DDR4_TEN
PS_DDR4_RESET_B
PS_DDR4_CS_B
PS_DDR4_A16_RAS_B
PS_DDR4_A15_CAS_B
PS_DDR4_BA1
PS_DDR4_BA0
PS_DDR4_CK_C
PS_DDR4_CK_T
PS_DDR4_ACT_B
PS_DDR4_A14_WE_B
PS_DDR4_ODT_B
PS_DDR4_CKE
PS_DDR4_PAR
PS_DDR4_A[0..13]
DDR4_VPP_2V5
DDR4_VTT_PS
PS_DDR4_A0
PS_DDR4_A1
PS_DDR4_A2
PS_DDR4_A3
PS_DDR4_A4
PS_DDR4_A5
PS_DDR4_A6
PS_DDR4_A7
PS_DDR4_A8
PS_DDR4_A9
PS_DDR4_A10
PS_DDR4_A11
PS_DDR4_A12
PS_DDR4_A13
PS_DDR4_A14_WE_B
PS_DDR4_A15_CAS_B
PS_DDR4_A16_RAS_B
PS_DDR4_BA0
PS_DDR4_BA1
PS_DDR4_BG0
PS_DDR4_CKE
PS_DDR4_ACT_B
PS_DDR4_PAR
PS_DDR4_ODT_B
PS_DDR4_CS_B
RM55
RM56
RM57
RM58
RM59
RM60
RM61
RM65
RM62
RM66
RM67
RM68
RM69
RM70
RM71
RM72
RM74
RM75
RM76
RM77
RM78
RM79
RM80
RM82
RM83
R,39.2R,1%
PS_DDR4_TEN
PS_DDR4_ALERT_B
PS_DDR4_RESET_B
RM84
RM85
RM86
VCC_1V2
R,499R,1%
R,4.75K,1%
R,1K,1%
PS_DDR4_CK_C
PS_DDR4_CK_T
RM88
RM87
CM113
0.01uF/25V/X7R/10%
R,36.5R,1%
R,36.5R,1%
VCC_1V2
RES,10K,0402
0.01uF/25V/X7R/10%
RM63
R,10K,1%
RM64
CM105
CM106
CM107
10uF/25V/X5R/10%
10uF/25V/X5R/10%
UM9
REFIN
VLDOIN
VO
VOSNS
VIN
PGOOD
EN
REFOUT
PGND
GND
PAD_GND
TPS51200DRCT
VCC_3V3
CM108
0.1uF/25V/X7R/10%
RM73
R,100K,1%
DDR4_VREF_PS
CM110
CM111
0.1uF/25V/X7R/10%
4.7uF/6.3V/X5R/20%
DDR4_VTT_PS
CM109
CM112
RM81
R,0R,1%
10uF/25V/X5R/10%
100uF/6.3V/X5R/20%
PLACE VOLTAGE SENSE POINT NEAR DDR4 COMPONENTS
DDR4_VTT_PS
DDR4_VTT_PS
CM114
CM115
CM116
CM117
CM118
CM119
CM120
CM121
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
0.1uF/25V/X7R/10%
VCC_3V3
UM10
IN1
IN2
IN3
EN
IC
OUT2
OUT1
FB
SS
GND
EP
MAX15027
DDR4_VPP_2V5
RM89
R,40.2K,1%
RM90
R,10K,1%
CM122
CM123
CM124
4.7uF/6.3V/X5R/20%
4.7uF/6.3V/X5R/20%
0.01uF/25V/X7R/10%
VOUT = 0.5V(1+RU/RD)
DDR4_1 VPP 2.5V
2.5V@MAX 1A
V3BEST
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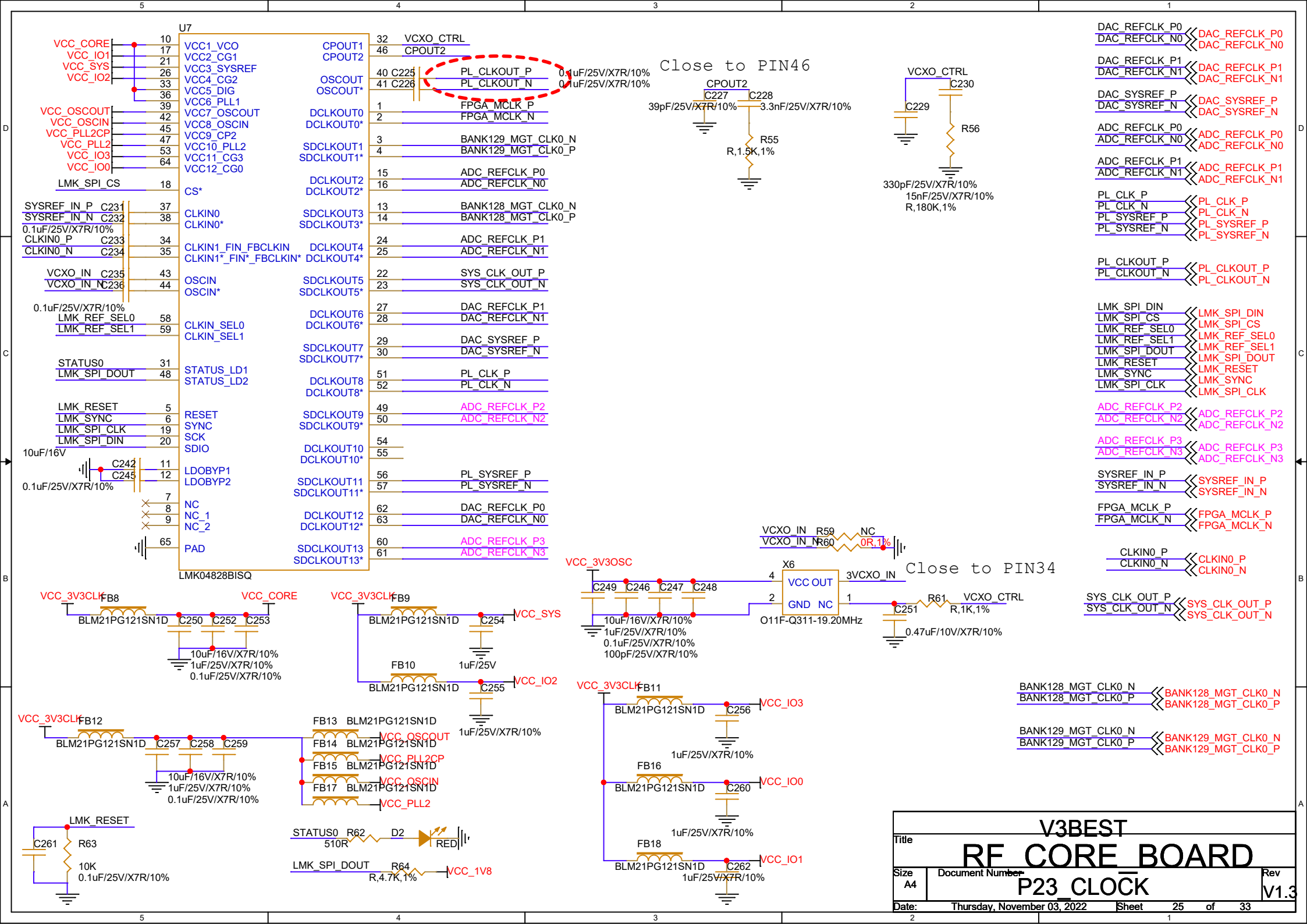


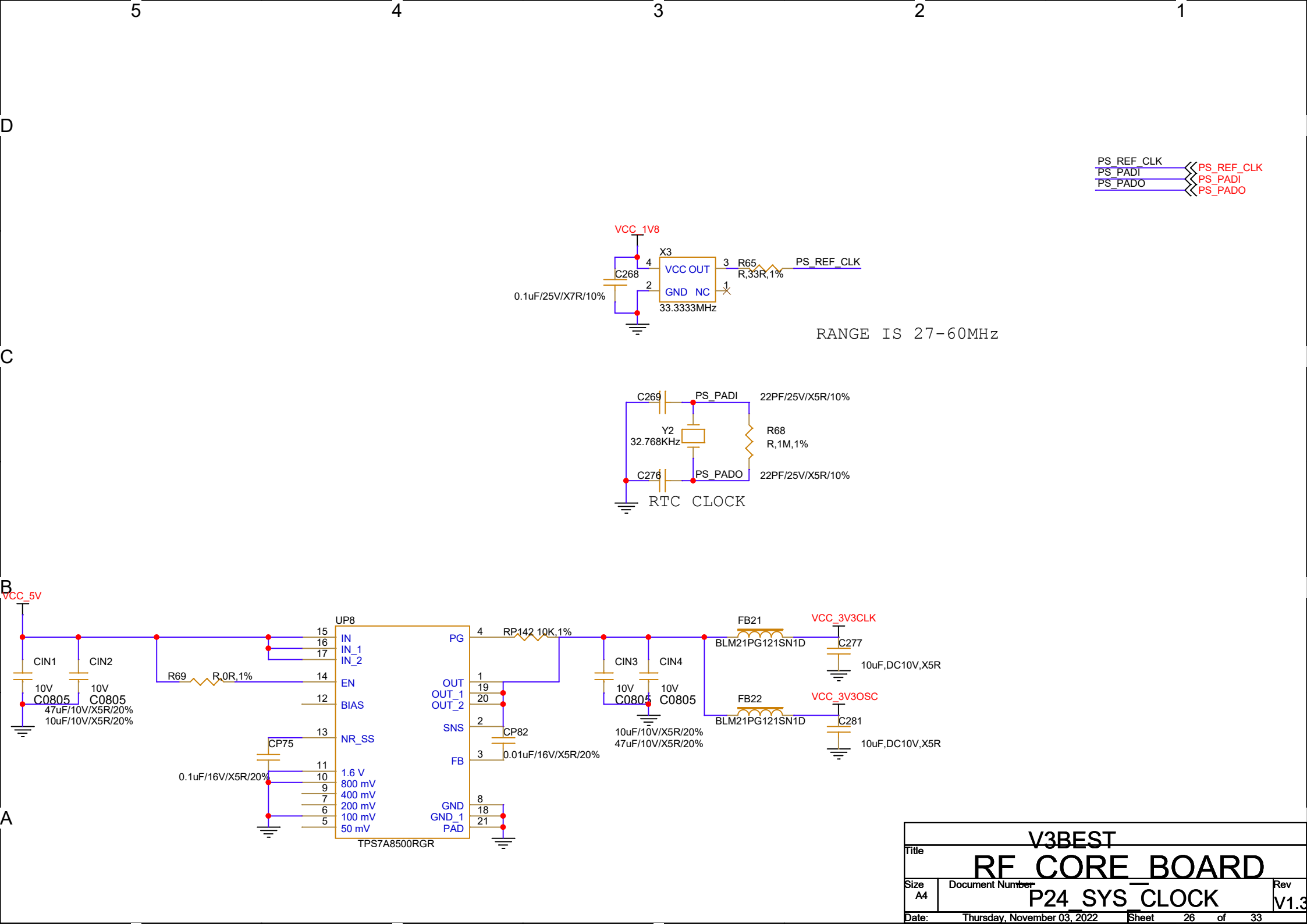
V3BEST

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U7
LMK04828BISQ
VCC_CORE
VCC_IO1
VCC_SYS
VCC_IO2
VCC_OSCOUT
VCC_OSCIN
VCC_PLL2CP
VCC_PLL2
VCC_IO3
VCC_IO0
LMK_SPI_CS
SYSREF_IN_P
SYSREF_IN_N
CLKIN0_P
CLKIN0_N
VCXO_IN
VCXO_IN_N
LMK_REF_SEL0
LMK_REF_SEL1
STATUS0
LMK_SPI_DOUT
LMK_RESET
LMK_SYNC
LMK_SPI_CLK
LMK_SPI_DIN
VCC1_VCO
VCC2_CG1
VCC3_SYSREF
VCC4_CG2
VCC5_DIG
VCC6_PLL1
VCC7_OSCOUT
VCC8_OSCIN
VCC9_CP2
VCC10_PLL2
VCC11_CG3
VCC12_CG0
CS*
CLKIN0
CLKIN0*
CLKIN1_FIN_FBCLKIN
CLKIN1*_FIN*_FBCLKIN*
OSCIN
OSCIN*
CLKIN_SEL0
CLKIN_SEL1
STATUS_LD1
STATUS_LD2
RESET
SYNC
SCK
SDIO
LDOBYP1
LDOBYP2
NC
NC_1
NC_2
PAD
CPOUT1
CPOUT2
OSCOUT
OSCOUT*
DCLKOUT0
DCLKOUT0*
SDCLKOUT1
SDCLKOUT1*
DCLKOUT2
DCLKOUT2*
SDCLKOUT3
SDCLKOUT3*
DCLKOUT4
DCLKOUT4*
SDCLKOUT5
SDCLKOUT5*
DCLKOUT6
DCLKOUT6*
SDCLKOUT7
SDCLKOUT7*
DCLKOUT8
DCLKOUT8*
SDCLKOUT9
SDCLKOUT9*
DCLKOUT10
DCLKOUT10*
SDCLKOUT11
SDCLKOUT11*
DCLKOUT12
DCLKOUT12*
SDCLKOUT13
SDCLKOUT13*
VCXO_CTRL
PL_CLKOUT_P
PL_CLKOUT_N
FPGA_MCLK_P
FPGA_MCLK_N
BANK129_MGT_CLK0_N
BANK129_MGT_CLK0_P
ADC_REFCLK_P0
ADC_REFCLK_N0
BANK128_MGT_CLK0_N
BANK128_MGT_CLK0_P
ADC_REFCLK_P1
ADC_REFCLK_N1
SYS_CLK_OUT_P
SYS_CLK_OUT_N
DAC_REFCLK_P1
DAC_REFCLK_N1
DAC_SYSREF_P
DAC_SYSREF_N
PL_CLK_P
PL_CLK_N
PL_SYSREF_P
PL_SYSREF_N
ADC_REFCLK_P2
ADC_REFCLK_N2
DAC_REFCLK_P0
DAC_REFCLK_N0
ADC_REFCLK_P3
ADC_REFCLK_N3
0.1uF/25V/X7R/10%
10uF/16V
C242
C245
C231
C232
C233
C234
C235
C236
C249
C246
C247
C248
C227
C228
C229
C230
R55
R,1.5K,1%
R56
330pF/25V/X7R/10%
15nF/25V/X7R/10%
R,180K,1%
Close to PIN46
39pF/25V/X7R/10%
3.3nF/25V/X7R/10%
C225
C226
0.1uF/25V/X7R/10%
VCC_3V3OSC
VCXO_IN
VCXO_IN_N
R59
R60
NC
0R,1%
X6
VCC OUT
GND NC
3VCXO_IN
O11F-Q311-19.20MHz
Close to PIN34
C251
R61
R,1K,1%
0.47uF/10V/X7R/10%
10uF/16V/X7R/10%
1uF/25V/X7R/10%
100pF/25V/X7R/10%
VCC_3V3CLK
FB8
FB9
FB10
FB11
FB12
FB13
FB14
FB15
FB16
FB17
FB18
BLM21PG121SN1D
C250
C252
C253
C254
C255
C256
C257
C258
C259
C260
C261
C262
1uF/25V
1uF/25V/X7R/10%
VCC_SYS
VCC_IO2
VCC_IO3
VCC_IO0
VCC_IO1
VCC_OSCOUT
VCC_PLL2CP
VCC_OSCIN
VCC_PLL2
LMK_RESET
R63
10K
STATUS0
R62
510R
D2
RED
LMK_SPI_DOUT
R64
R,4.7K,1%
VCC_1V8
V3BEST
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U11

VCC_5V

22uF/16V/X5R/20%
22uF/16V/X5R/20%
22uF/16V/X5R/20%

CIN5 16V
CIN6 16V C1206
CIN7 16V C1206
CIN8 16V C1206
CIN9 16V C1206

100uF/16V/X5R/20%
22uF/16V/X5R/20%

VCC_5V

RP7 130K 1%

VCC_5V

C283 0.1uF/25V/X7R/10%

INTVCC RP156 10R,5%

INTVCC

22uF/6.3V/X5R/20% CIN10 10V C1206

CIN11 4.7uF/6.3V/X5R/20%

C287 NC

K4 VIN1A
K5 VIN1B
L4 VIN1C
L5 VIN1D
A4 VIN1E
A5 VIN1F
B4 VIN1G
B5 VIN1H

F8 CPWR

H5 RUN1
D5 RUN2

H4 TRACK/SS1
F2 TRACK/SS2

F3 MODE-PLLIN

F4 CLK_OUT

F6 INTVCC

G7 DRVCC

E7 EXTVCC

H2 PHASEMD

D3 SGND1
H3 SGND2

A3 GND1
A6 GND2
A7 GND3
B3 GND4
B6 GND5
B7 GND6
C3 GND7
C4 GND8
C5 GND9
C6 GND10
C7 GND11
D6 GND12
D7 GND13
E6 GND14
E8 GND15
F5 GND16
F7 GND17
G6 GND18
G8 GND19
H6 GND20
H7 GND21
J4 GND22
J5 GND23
J6 GND24
J7 GND25
K3 GND26
K6 GND27
K7 GND28
L3 GND29
L6 GND30
L7 GND31

POWER BY LINEAR™

H1 VOUT1A
J1 VOUT1B
J2 VOUT1C
K1 VOUT1D
K2 VOUT1E
L1 VOUT1F
L2 VOUT1G

TP11 VCCINT_0V72

COUT1 4V
COUT2 4V
COUT3 4V

100uF/4V/X5R/20%
100uF/4V/X5R/20%
100uF/4V/X5R/20%

15A

G2 VOUTS1

G4 VFB1 — INTVCC

J3 VOUTS1-

H8 SW1_0
J8 SW1_1

G5 PGOOD1

G3 COMP1A — COMP1A

G1 COMP1B

D2 VRNG — INTVCC

F1 FREQ — RP17 39.2K 1%

L8 TMP1+

K8 TMP1-

PGOOD0 ≪ PGOOD0

A1 VOUT2A
A2 VOUT2B
B1 VOUT2C
B2 VOUT2D
C1 VOUT2E
C2 VOUT2F
D1 VOUT2G

VCCINT_0V72

COUT4 6.3V
COUT5 6.3V
COUT6 6.3V

100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%

15A

E2 VOUTS2 — RP23 0.1%

E4 VFB2 — RP25 143K,1% / RP26 41.2K 1%

D4 VOUTS2-

C8 SW2_0

D8 SW2_1

E5 PGOOD2 — RP29 10K 1% — INTVCC / PGOOD0

E3 COMP2A — COMP1A — C289 100pF/6.3V/X5R/20% — RP31 7.15K,1% — C637 1500pF/6.3V/X5R/20%

E1 COMP2B

B8 TEMP2+

A8 TEMP2-

INTVCC — RP22 510 — DDP2 GREEN

QP2 2N7002A-7 — PGOOD0

应该使用两路LTM4662并联输出30A电流模式

VOUT1=0.6V*((60.4K+RFB)/RFB)

VOUT2=0.6V*((60.4K+RFB2)/RFB2)

RFB3=(60.4K*RFB2)/(RFB2+60.4K)

FREQ	115kΩ	350KHz
FREQ	165kΩ	250KHz
FREQ	39.2kΩ	1000KHz

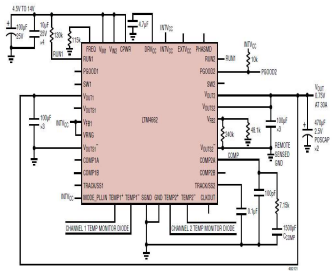
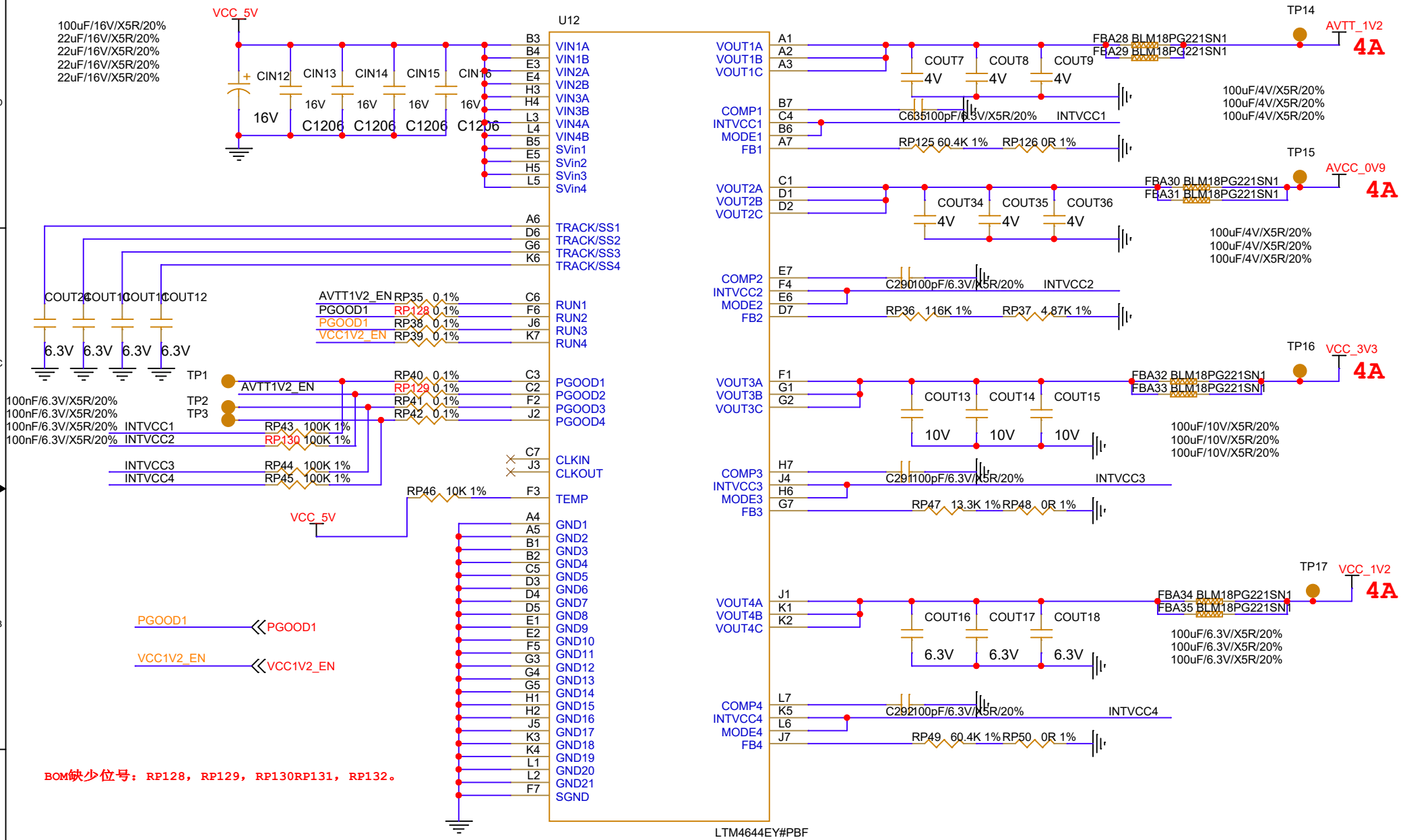


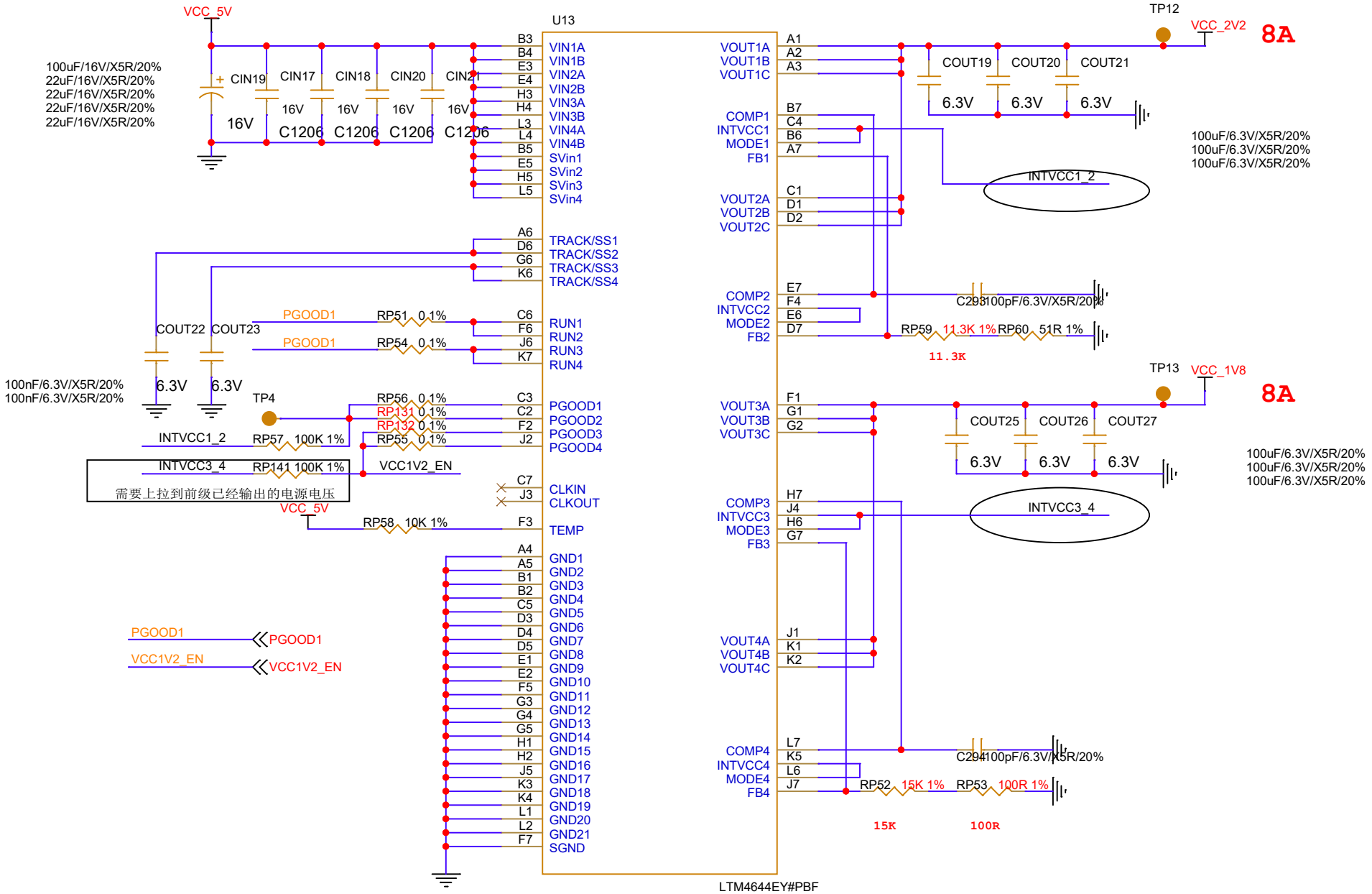
Figure 21. 12V Input to 0.75V at 30A Two Phase

LTM4662IY#PBF

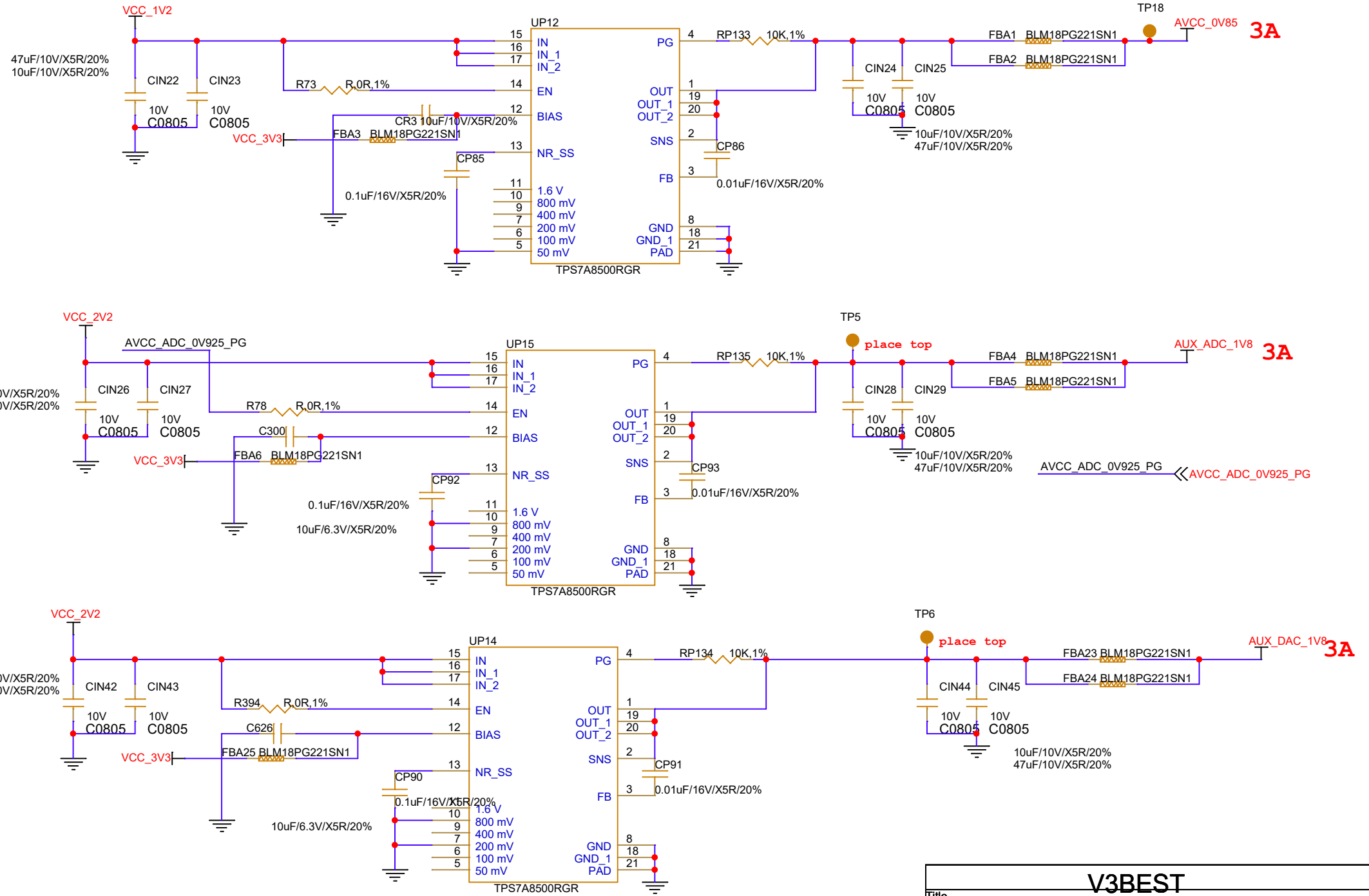
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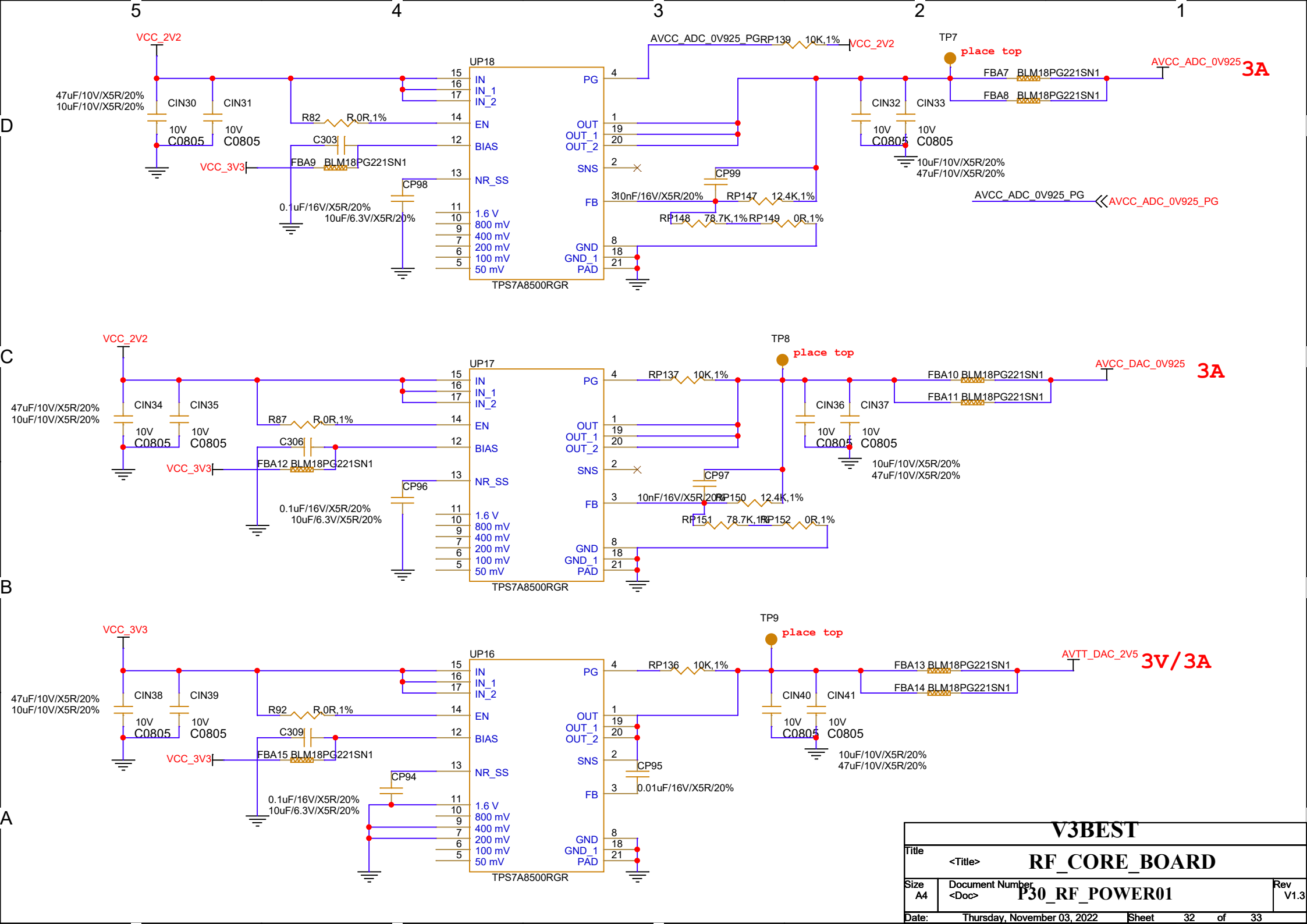
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5 4 3 2 1

D C B A

VCC_5V

22uF/16V/X5R/20%
22uF/16V/X5R/20%
22uF/16V/X5R/20%

CIN46 CIN47 CIN48 CIN49 CIN50
16V 16V 16V 16V 16V
16V C1206 C1206 C1206 C1206

100uF/16V/X5R/20%
22uF/16V/X5R/20%

VCC_5V

PGOOD0 RP98 R,130K 1%

C628 0.1uF/25V/X7R/10%

INTVCC_00 RP155 10R,5%

INTVCC_00

CIN51 22uF/6.3V/X5R/20% 10V C1206

CIN52 4.7uF/6.3V/X5R/20%

C632 NC

U107

K4 K5 L4 L5 A4 A5 B4 B5 — VIN1A VIN1B VIN1C VIN1D VIN1E VIN1F VIN1G VIN1H

F8 — CPWR

H5 — RUN1
D5 — RUN2

H4 — TRACK/SS1
F2 — TRACK/SS2

F3 — MODE-PLLIN

F4 — CLK_OUT

F6 — INTVCC

G7 — DRVCC

E7 — EXTVCC

H2 — PHASEMD

D3 H3 — SGND1 SGND2

A3 A6 A7 B3 B6 B7 C3 C4 C5 C6 C7 D6 D7 E6 E8 F5 F7 G6 G8 H6 H7 J4 J5 J6 J7 K3 K6 K7 L3 L6 L7 — GND1 GND2 GND3 GND4 GND5 GND6 GND7 GND8 GND9 GND10 GND11 GND12 GND13 GND14 GND15 GND16 GND17 GND18 GND19 GND20 GND21 GND22 GND23 GND24 GND25 GND26 GND27 GND28 GND29 GND30 GND31

H1 J1 J2 K1 K2 L1 L2 — VOUT1A VOUT1B VOUT1C VOUT1D VOUT1E VOUT1F VOUT1G

TP10 VCCINT_0V85

COUT28 COUT29 COUT30
4V 4V 4V

100uF/4V/X5R/20%
100uF/4V/X5R/20%
100uF/4V/X5R/20%

15A

G2 — VOUTS1

G4 — VFB1 — INTVCC_00

J3 — VOUTS1-

H8 J8 — SW1_0 SW1_1

G5 — PGOOD1

G3 — COMP1A — COMP2A

G1 — COMP1B

D2 — VRNG — INTVCC_00

F1 — FREQ — RP107 39.2K 1%

L8 — TMP1+

K8 — TMP1-

A1 A2 B1 B2 C1 C2 D1 — VOUT2A VOUT2B VOUT2C VOUT2D VOUT2E VOUT2F VOUT2G

VCCINT_0V85

COUT31 COUT32 COUT33
6.3V 6.3V 6.3V

100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%
100uF/6.3V/X5R/20%

15A

E2 — VOUTS2 — RP113 R,0 1%

E4 — VFB2 — RP115 143K,1% RP116 41.2K 1%

D4 — VOUTS2-

C8 — SW2_0

D8 — SW2_1

E5 — PGOOD2 — RP119 R,10K 1% — INTVCC_00 PGOOD1

E3 — COMP2A — C634 100pF/6.3V/X5R/20% — RP121 7.15K,1% — C636 1500pF/6.3V/X5R/20%

E1 — COMP2B

B8 — TEMP2+

A8 — TEMP2-

INTVCC_00 — RP112 R,510R,1%

DDP4 GREEN

QP4 2N7002A-7

PGOOD0 — PGOOD0
PGOOD1 — PGOOD1

应该使用两路LTM4662并联输出30A电流模式

LTM4662IY#PBF

POWER BY LINEAR™

VOUT1=0.6V*((60.4K+RFB)/RFB)

VOUT2=0.6V*((60.4K+RFB2)/RFB2)

RFB3=(60.4K*RFB2)/(RFB2+60.4K)

FREQ	115kΩ	350KHz
FREQ	165kΩ	250KHz
FREQ	39.2kΩ	1000KHz

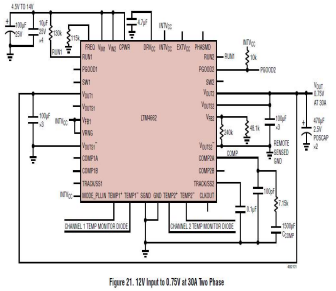
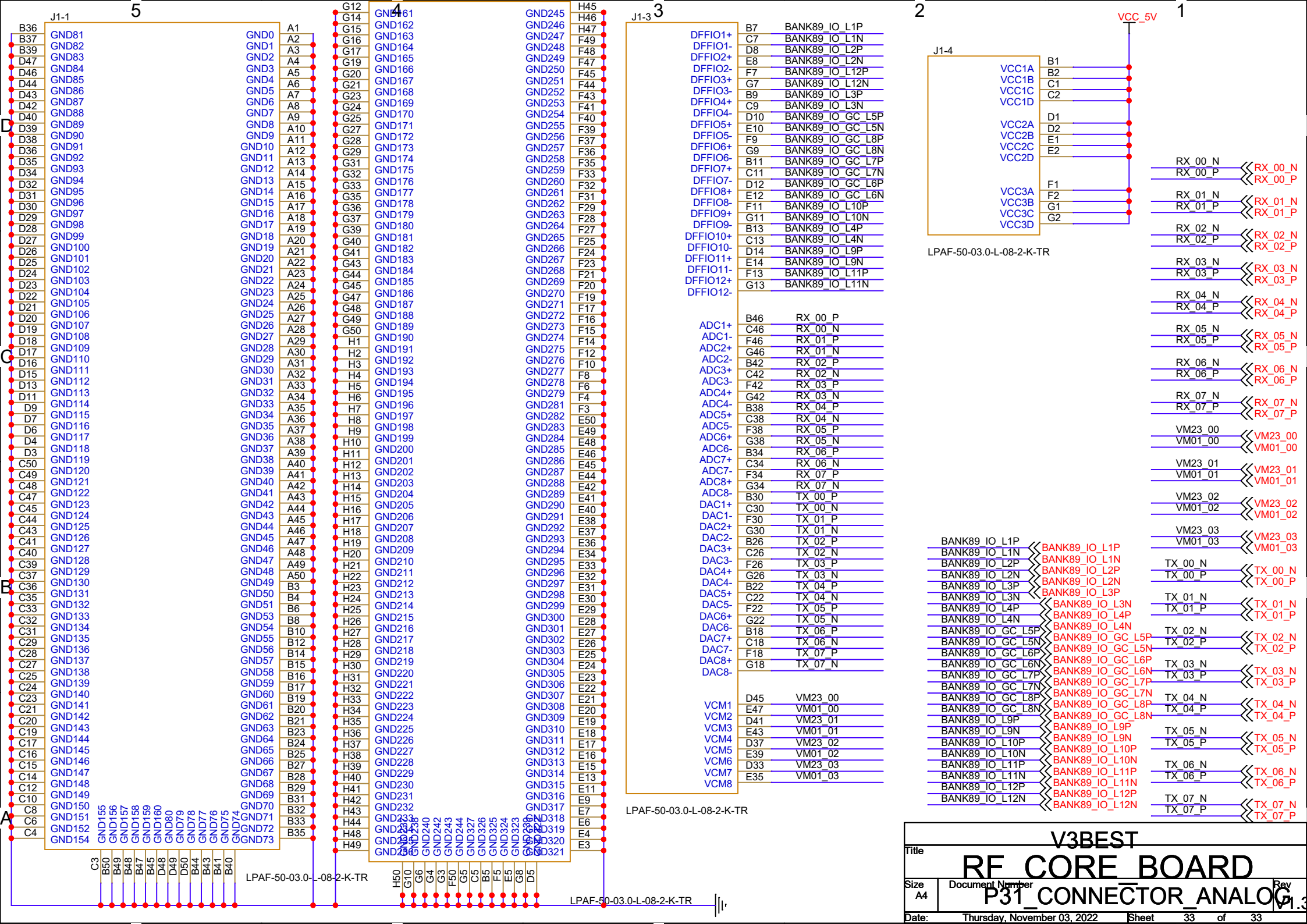
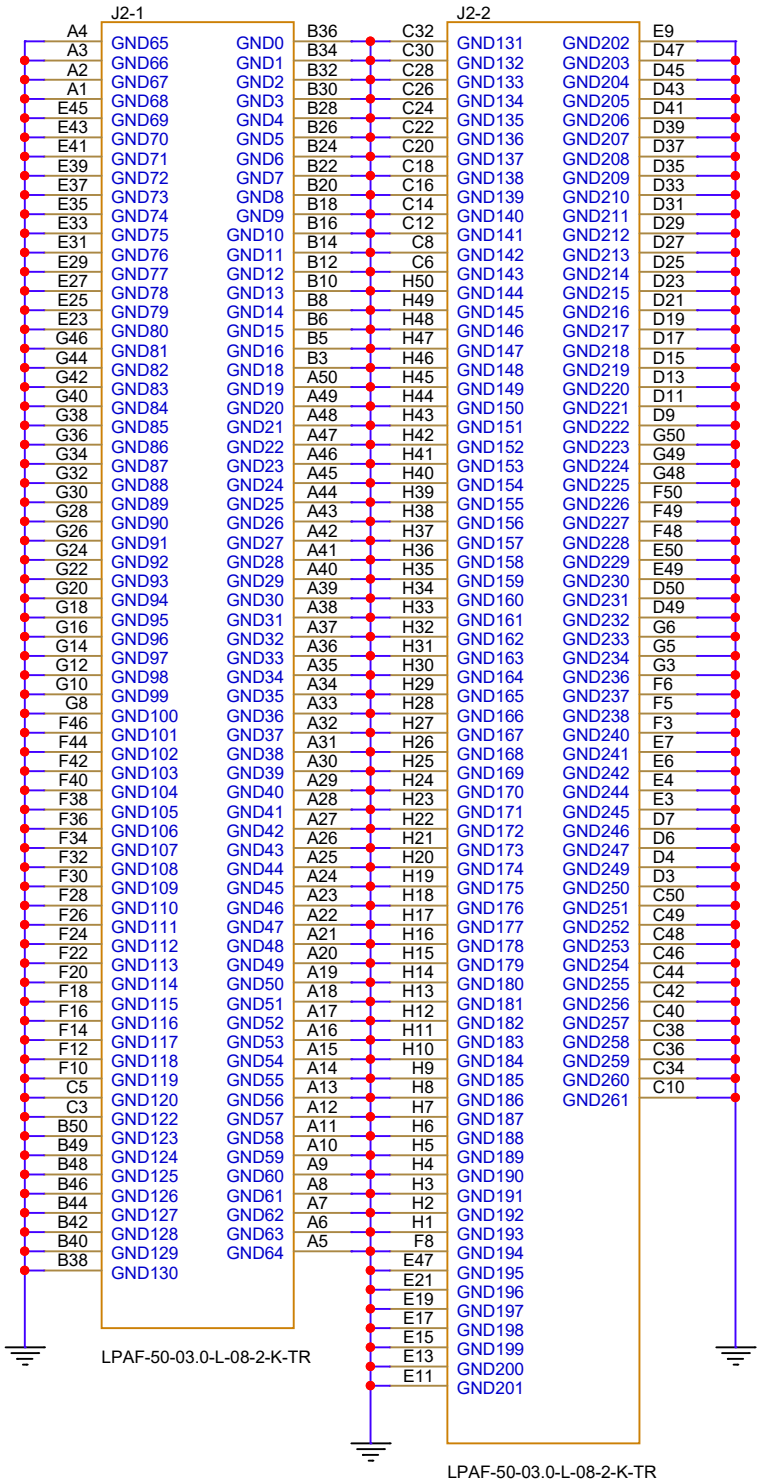


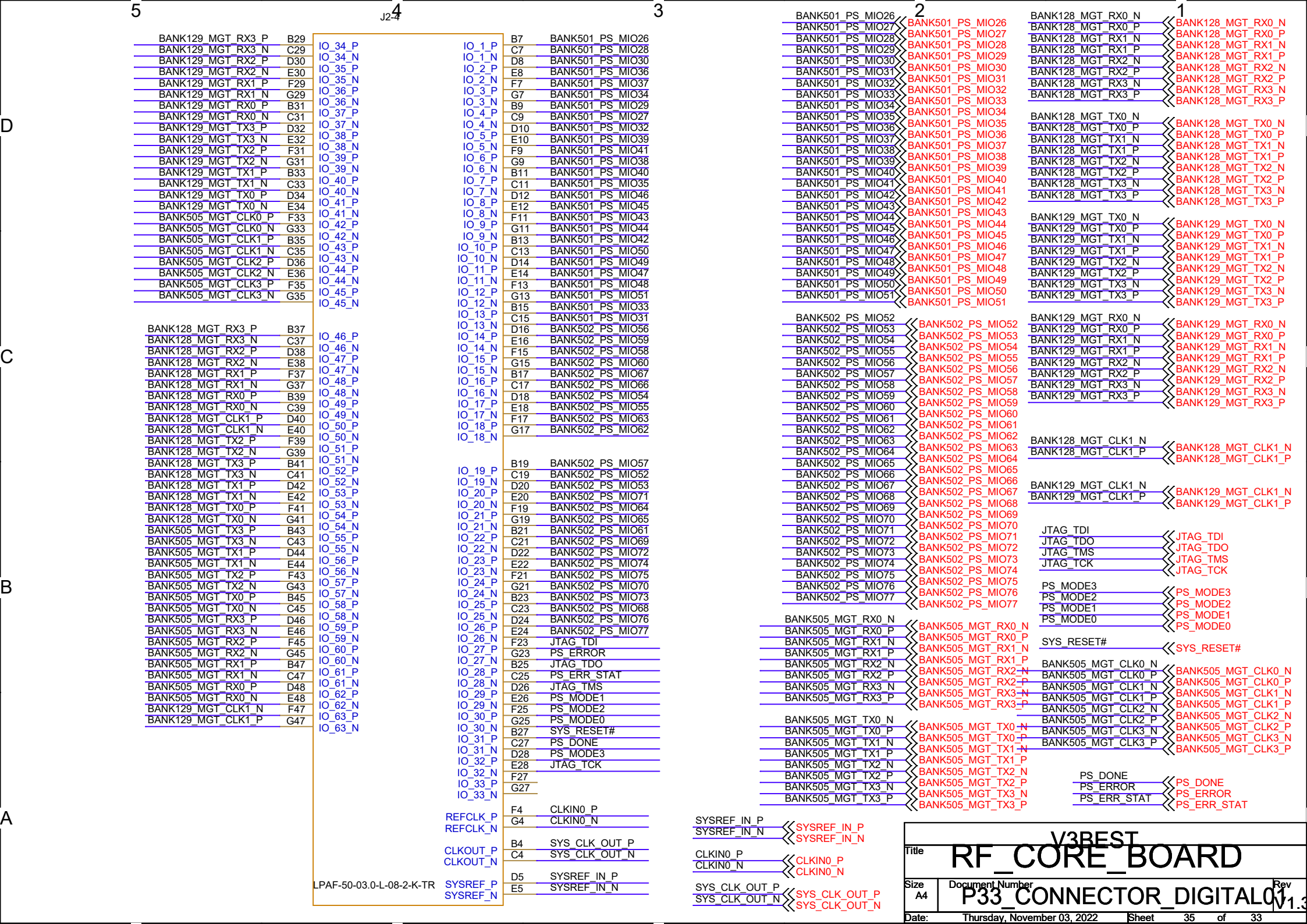
Figure 21. 12V Input to 0.75V at 30A Two Phase

Title	V3BEST ZING7		
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J1-1
J1-3
J1-4
LPAF-50-03.0-L-08-2-K-TR
VCC_5V
VCC1A
VCC1B
VCC1C
VCC1D
VCC2A
VCC2B
VCC2C
VCC2D
VCC3A
VCC3B
VCC3C
VCC3D
DFFIO1+
DFFIO1-
DFFIO2+
DFFIO2-
DFFIO3+
DFFIO3-
DFFIO4+
DFFIO4-
DFFIO5+
DFFIO5-
DFFIO6+
DFFIO6-
DFFIO7+
DFFIO7-
DFFIO8+
DFFIO8-
DFFIO9+
DFFIO9-
DFFIO10+
DFFIO10-
DFFIO11+
DFFIO11-
DFFIO12+
DFFIO12-
BANK89_IO_L1P
BANK89_IO_L1N
BANK89_IO_L2P
BANK89_IO_L2N
BANK89_IO_L12P
BANK89_IO_L12N
BANK89_IO_L3P
BANK89_IO_L3N
BANK89_IO_GC_L5P
BANK89_IO_GC_L5N
BANK89_IO_GC_L8P
BANK89_IO_GC_L8N
BANK89_IO_GC_L7P
BANK89_IO_GC_L7N
BANK89_IO_GC_L6P
BANK89_IO_GC_L6N
BANK89_IO_L10P
BANK89_IO_L10N
BANK89_IO_L4P
BANK89_IO_L4N
BANK89_IO_L9P
BANK89_IO_L9N
BANK89_IO_L11P
BANK89_IO_L11N
ADC1+
ADC1-
ADC2+
ADC2-
ADC3+
ADC3-
ADC4+
ADC4-
ADC5+
ADC5-
ADC6+
ADC6-
ADC7+
ADC7-
ADC8+
ADC8-
DAC1+
DAC1-
DAC2+
DAC2-
DAC3+
DAC3-
DAC4+
DAC4-
DAC5+
DAC5-
DAC6+
DAC6-
DAC7+
DAC7-
DAC8+
DAC8-
RX_00_P
RX_00_N
RX_01_P
RX_01_N
RX_02_P
RX_02_N
RX_03_P
RX_03_N
RX_04_P
RX_04_N
RX_05_P
RX_05_N
RX_06_P
RX_06_N
RX_07_P
RX_07_N
TX_00_P
TX_00_N
TX_01_P
TX_01_N
TX_02_P
TX_02_N
TX_03_P
TX_03_N
TX_04_P
TX_04_N
TX_05_P
TX_05_N
TX_06_P
TX_06_N
TX_07_P
TX_07_N
VCM1
VCM2
VCM3
VCM4
VCM5
VCM6
VCM7
VCM8
VM23_00
VM01_00
VM23_01
VM01_01
VM23_02
VM01_02
VM23_03
VM01_03
V3BEST
RF_CORE_BOARD
P31_CONNECTOR_ANALOG
Size A4
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4
3
2
1
D
C
B
A
J2-4
LPAF-50-03.0-L-08-2-K-TR
BANK129_MGT_RX3_P B29
BANK129_MGT_RX3_N C29
BANK129_MGT_RX2_P D30
BANK129_MGT_RX2_N E30
BANK129_MGT_RX1_P F29
BANK129_MGT_RX1_N G29
BANK129_MGT_RX0_P B31
BANK129_MGT_RX0_N C31
BANK129_MGT_TX3_P D32
BANK129_MGT_TX3_N E32
BANK129_MGT_TX2_P F31
BANK129_MGT_TX2_N G31
BANK129_MGT_TX1_P B33
BANK129_MGT_TX1_N C33
BANK129_MGT_TX0_P D34
BANK129_MGT_TX0_N E34
BANK505_MGT_CLK0_P F33
BANK505_MGT_CLK0_N G33
BANK505_MGT_CLK1_P B35
BANK505_MGT_CLK1_N C35
BANK505_MGT_CLK2_P D36
BANK505_MGT_CLK2_N E36
BANK505_MGT_CLK3_P F35
BANK505_MGT_CLK3_N G35
IO_34_P
IO_34_N
IO_35_P
IO_35_N
IO_36_P
IO_36_N
IO_37_P
IO_37_N
IO_38_P
IO_38_N
IO_39_P
IO_39_N
IO_40_P
IO_40_N
IO_41_P
IO_41_N
IO_42_P
IO_42_N
IO_43_P
IO_43_N
IO_44_P
IO_44_N
IO_45_P
IO_45_N
BANK128_MGT_RX3_P B37
BANK128_MGT_RX3_N C37
BANK128_MGT_RX2_P D38
BANK128_MGT_RX2_N E38
BANK128_MGT_RX1_P F37
BANK128_MGT_RX1_N G37
BANK128_MGT_RX0_P B39
BANK128_MGT_RX0_N C39
BANK128_MGT_CLK1_P D40
BANK128_MGT_CLK1_N E40
BANK128_MGT_TX2_P F39
BANK128_MGT_TX2_N G39
BANK128_MGT_TX3_P B41
BANK128_MGT_TX3_N C41
BANK128_MGT_TX1_P D42
BANK128_MGT_TX1_N E42
BANK128_MGT_TX0_P F41
BANK128_MGT_TX0_N G41
BANK505_MGT_TX3_P B43
BANK505_MGT_TX3_N C43
BANK505_MGT_TX1_P D44
BANK505_MGT_TX1_N E44
BANK505_MGT_TX2_P F43
BANK505_MGT_TX2_N G43
BANK505_MGT_TX0_P B45
BANK505_MGT_TX0_N C45
BANK505_MGT_RX3_P D46
BANK505_MGT_RX3_N E46
BANK505_MGT_RX2_P F45
BANK505_MGT_RX2_N G45
BANK505_MGT_RX1_P B47
BANK505_MGT_RX1_N C47
BANK505_MGT_RX0_P D48
BANK505_MGT_RX0_N E48
BANK129_MGT_CLK1_N F47
BANK129_MGT_CLK1_P G47
IO_46_P
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IO_47_P
IO_47_N
IO_48_P
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IO_29_P
IO_29_N
IO_30_P
IO_30_N
IO_31_P
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IO_32_P
IO_32_N
IO_33_P
IO_33_N
REFCLK_P
REFCLK_N
CLKOUT_P
CLKOUT_N
SYSREF_P
SYSREF_N
B7 BANK501_PS_MIO26
C7 BANK501_PS_MIO28
D8 BANK501_PS_MIO30
E8 BANK501_PS_MIO36
F7 BANK501_PS_MIO37
G7 BANK501_PS_MIO34
B9 BANK501_PS_MIO29
C9 BANK501_PS_MIO27
D10 BANK501_PS_MIO32
E10 BANK501_PS_MIO39
F9 BANK501_PS_MIO41
G9 BANK501_PS_MIO38
B11 BANK501_PS_MIO40
C11 BANK501_PS_MIO35
D12 BANK501_PS_MIO46
E12 BANK501_PS_MIO45
F11 BANK501_PS_MIO43
G11 BANK501_PS_MIO44
B13 BANK501_PS_MIO42
C13 BANK501_PS_MIO50
D14 BANK501_PS_MIO49
E14 BANK501_PS_MIO47
F13 BANK501_PS_MIO48
G13 BANK501_PS_MIO51
B15 BANK501_PS_MIO33
C15 BANK501_PS_MIO31
D16 BANK502_PS_MIO56
E16 BANK502_PS_MIO59
F15 BANK502_PS_MIO58
G15 BANK502_PS_MIO60
B17 BANK502_PS_MIO67
C17 BANK502_PS_MIO66
D18 BANK502_PS_MIO54
E18 BANK502_PS_MIO55
F17 BANK502_PS_MIO63
G17 BANK502_PS_MIO62
B19 BANK502_PS_MIO57
C19 BANK502_PS_MIO52
D20 BANK502_PS_MIO53
E20 BANK502_PS_MIO71
F19 BANK502_PS_MIO64
G19 BANK502_PS_MIO65
B21 BANK502_PS_MIO61
C21 BANK502_PS_MIO69
D22 BANK502_PS_MIO72
E22 BANK502_PS_MIO74
F21 BANK502_PS_MIO75
G21 BANK502_PS_MIO70
B23 BANK502_PS_MIO73
C23 BANK502_PS_MIO68
D24 BANK502_PS_MIO76
E24 BANK502_PS_MIO77
F23 JTAG_TDI
G23 PS_ERROR
B25 JTAG_TDO
C25 PS_ERR_STAT
D26 JTAG_TMS
E26 PS_MODE1
F25 PS_MODE2
G25 PS_MODE0
B27 SYS_RESET#
C27 PS_DONE
D28 PS_MODE3
E28 JTAG_TCK
F27
G27
F4 CLKIN0_P
G4 CLKIN0_N
B4 SYS_CLK_OUT_P
C4 SYS_CLK_OUT_N
D5 SYSREF_IN_P
E5 SYSREF_IN_N
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BANK501_PS_MIO28
BANK501_PS_MIO29
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BANK129_MGT_TX2_P
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BANK129_MGT_CLK1_P
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JTAG_TDO
JTAG_TMS
JTAG_TCK
PS_MODE3
PS_MODE2
PS_MODE1
PS_MODE0
SYS_RESET#
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BANK505_MGT_RX1_P
BANK505_MGT_RX2_N
BANK505_MGT_RX2_P
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BANK505_MGT_RX3_P
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BANK505_MGT_TX1_P
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BANK505_MGT_TX2_P
BANK505_MGT_TX3_N
BANK505_MGT_TX3_P
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BANK505_MGT_CLK0_P
BANK505_MGT_CLK1_N
BANK505_MGT_CLK1_P
BANK505_MGT_CLK2_N
BANK505_MGT_CLK2_P
BANK505_MGT_CLK3_N
BANK505_MGT_CLK3_P
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PS_ERROR
PS_ERR_STAT
SYSREF_IN_P
SYSREF_IN_N
CLKIN0_P
CLKIN0_N
SYS_CLK_OUT_P
SYS_CLK_OUT_N
V3REST
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RF_CORE_BOARD
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A4
Document Number
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